



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
 - Collected SpaceX launch data using SpaceX API (107 records) and web scraping from Wikipedia
 - Performed data wrangling to clean missing values and create landing success labels (Class: 1=success, 0=failure)
 - Conducted Exploratory Data Analysis (EDA) using Python visualizations and SQL queries
 - Built interactive maps with Folium to visualize launch sites and landing outcomes
 - Created a Plotly Dash dashboard with pie charts and scatter plots for interactive analysis
 - Developed and compared four classification models: Logistic Regression, SVM, Decision Tree, and KNN using GridSearchCV

Executive Summary

Summary of all results

- Best performing model: Logistic Regression with 83.33% test accuracy
- Most successful launch site: KSC LC-39A (highest success rate)
- Most successful payload range: 2,000–4,000 kg
- Best booster version: FT (almost 100% success rate)
- First successful ground pad landing: December 22, 2015
- Total successful missions: 60 out of 90 launches (66.7% success rate)

Introduction

Project background and context

SpaceX advertises Falcon 9 rocket launches at \$62 million, significantly lower than competitors' \$165 million. The primary cost saving comes from reusing the first stage booster. By predicting whether the first stage will land successfully, we can estimate launch costs and help alternative companies make competitive bids against SpaceX.

Introduction

Problems you want to find answers

1. Which launch sites have the highest success rates?
2. What payload mass ranges are most likely to succeed?
3. Which booster versions perform best?
4. How do orbit types affect landing success?
5. What is the relationship between launch outcomes and geographical locations?
6. Can we build an accurate machine learning model to predict landing success?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

Two methods were used to collect SpaceX launch data:

Method 1: SpaceX API

Used `requests.get()` to call the SpaceX public API endpoint: <https://api.spacexdata.com/v4/launches/past>

Retrieved JSON data containing 107 launch records

Extracted information from nested fields: rocket, payloads, launchpad, and cores using helper functions

Method 2: Web Scraping (Wikipedia)

Used BeautifulSoup to scrape https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

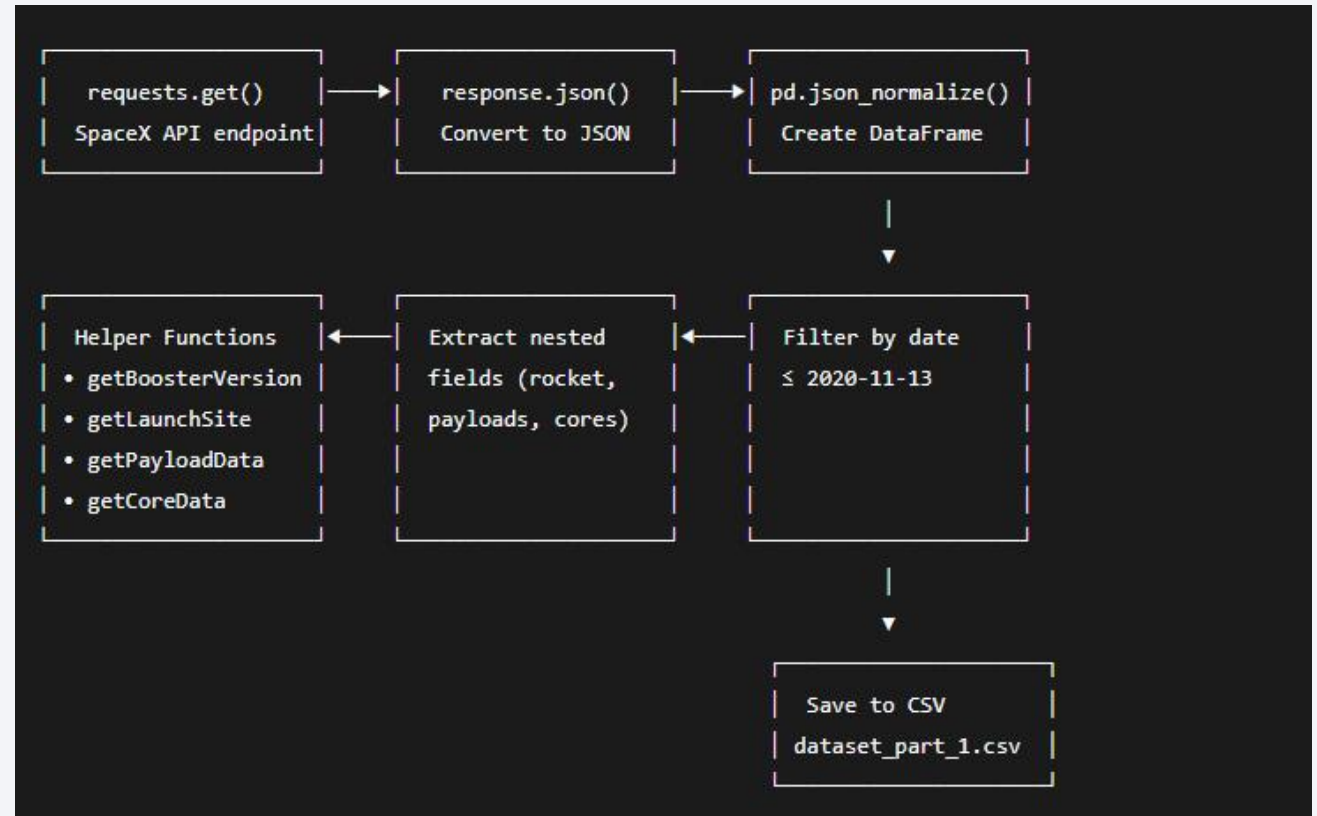
Parsed HTML tables to extract launch records including Flight Number, Date, Booster Version, Launch Site, Payload, Orbit, Customer, Outcome, and Landing status

Data Collection – SpaceX API

The data collection process began by using the `requests.get()` method to call the SpaceX public API endpoint `https://api.spacexdata.com/v4/launches/past`, which returned a JSON response containing 107 past launch records. The `json_normalize()` function was then used to convert the nested JSON data into a structured Pandas DataFrame. To extract specific nested information, several helper functions were implemented: `getBoosterVersion()` to retrieve booster names from the rocket field, `getLaunchSite()` to obtain launch site names and coordinates from the launchpad field, `getPayloadData()` to extract payload mass and orbit information, and `getCoreData()` to capture core details such as landing outcomes, reuse counts, and serial numbers. The data was then filtered to include only launches that occurred on or before November 13, 2020. Finally, the cleaned and processed dataset was saved as `dataset_part_1.csv`.

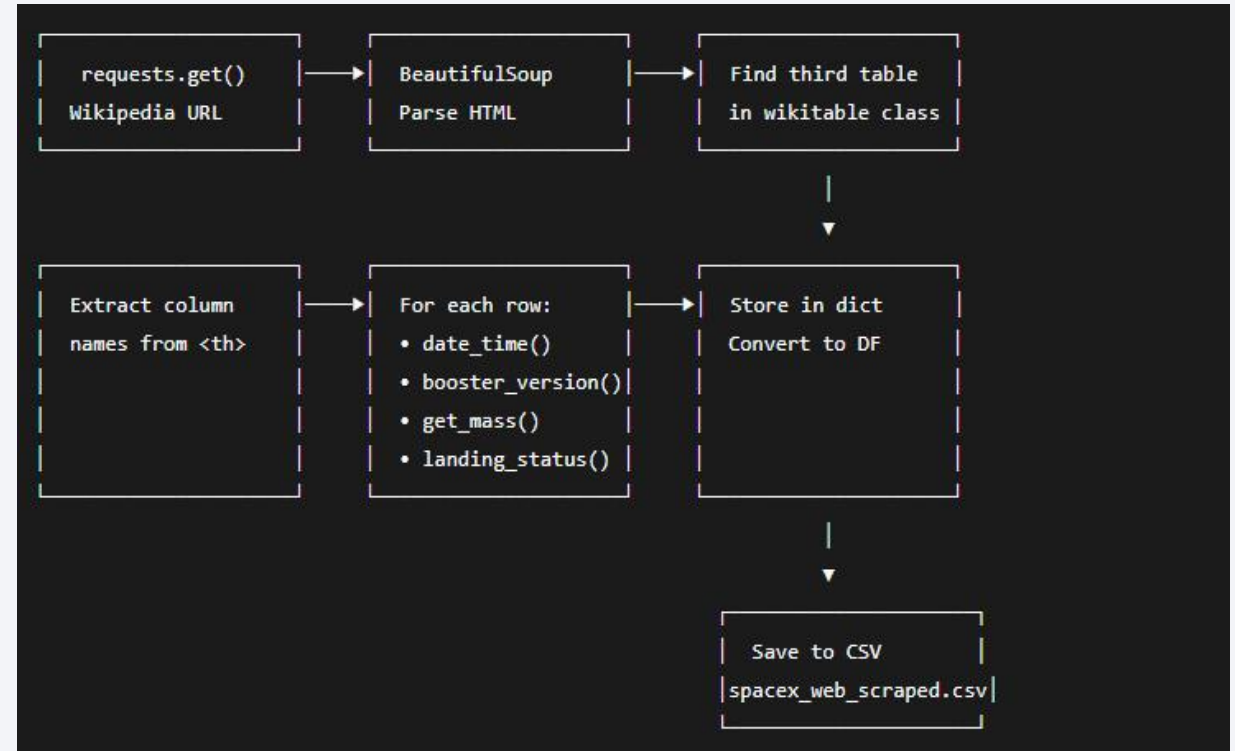
GitHub URL:

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>



Data Collection - Scraping

The web scraping process began by using the `requests.get()` method with custom headers to access the Wikipedia page "List of Falcon 9 and Falcon Heavy Launches" at the specified URL. The HTML content was then parsed using the BeautifulSoup library. The target table, which is the third table on the page containing the actual launch records, was located using the `find_all()` function. Column names were extracted from the `<th>` elements using a helper function called `extract_column_from_header()`. For each row in the table, the data was parsed using several helper functions: `date_time()` to extract date and time, `booster_version()` to get the booster version, `get_mass()` to retrieve payload mass, and `landing_status()` to capture the landing outcome. All extracted data was stored in a dictionary and then converted into a Pandas DataFrame. Finally, the cleaned dataset was saved as `spacex_web_scraped.csv`.



GitHub URL

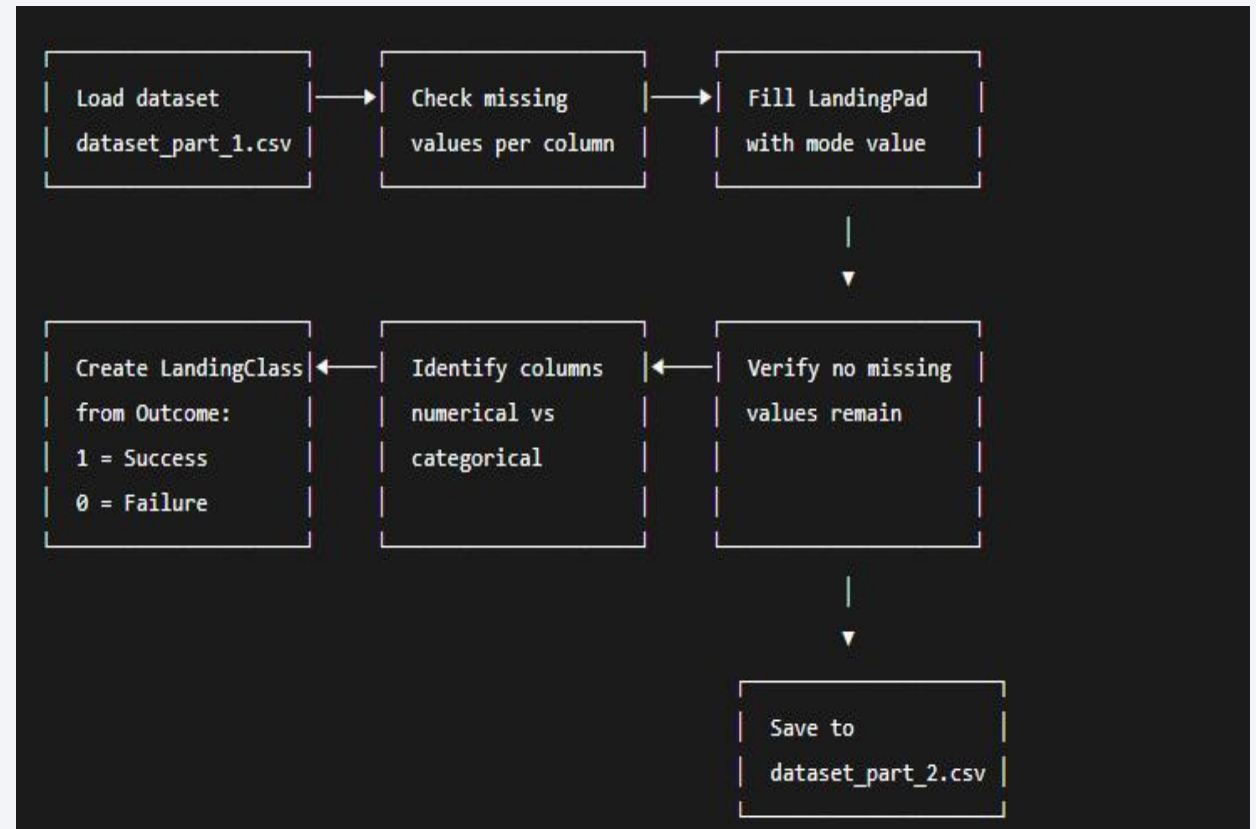
<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/jupyter-labs-webscraping.ipynb>

Data Wrangling

The data wrangling process began by loading the `dataset_part_1.csv` file into a Pandas DataFrame. First, missing values were identified by calculating the percentage of nulls in each column, revealing that the `LandingPad` column had approximately 28.9% missing values. These missing values were filled using the mode (most frequent value) of the `LandingPad` column to maintain data integrity. Next, the data types of all columns were examined to distinguish between numerical columns (`FlightNumber`, `PayloadMass`, `Flights`, `Block`, `ReusedCount`, `Longitude`, `Latitude`) and categorical columns (`Date`, `BoosterVersion`, `Orbit`, `LaunchSite`, `Outcome`, `Serial`, `LandingPad`). A new binary column called `LandingClass` was then created from the `Outcome` column, where 1 represented a successful landing (outcomes like "True ASDS", "True RTLS", "True Ocean") and 0 represented an unsuccessful landing (outcomes like "False ASDS", "False Ocean", "False RTLS", "None ASDS", "None None"). This transformed the prediction problem into a binary classification task. Finally, the cleaned dataset was saved as `dataset_part_2.csv`.

GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/jupyter-labs-webscraping.ipynb>



EDA with Data Visualization

Charts Plotted and Why They Were Used?

Chart Type	Purpose	Notebook Location
Launch Site Distribution	Show number of launches per site	Cell 11
Orbit Type Distribution	Analyze frequency of orbit destinations	Cell 13-15
Landing Outcome Distribution	Understand success vs failure patterns	Cell 16-18
Success Rate	Calculate overall landing percentage	Cell 25
Key Findings	GitHub URL	
• CCAFS SLC 40: 55 launches (most active) • KSC LC 39A: 22 launches • VAFB SLC 4E: 13 launches • GTO and ISS are most common orbit types • Overall success rate: 66.67%	https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb	

EDA with SQL

Using bullet point format, summarize the SQL queries you performed:

- 1- Displayed unique launch site names from the SPACEXTABLE.
- 2- Retrieved 5 records where launch sites begin with "CCA".
- 3- Calculated total payload mass carried by NASA (CRS) missions.
- 4- Computed average payload mass for booster version F9 v1.1.
- 5- Found the date of first successful landing outcome on ground pad.
- 6- Listed booster versions with successful drone ship landing and payload mass between 4000-6000 kg.
- 7- Counted total number of successful and failed mission outcomes.
- 8- Listed booster versions that carried the maximum payload mass using subquery.
- 9- Displayed month names, landing outcomes, booster versions, and launch sites for failed drone ship landings in 2015.
- 10 Ranked landing outcomes count between 2010-06-04 and 2017-03-20 in descending order.

GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

EDA with SQL

Key SQL Results:

- 1- Unique launch sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
- 2- Total payload for NASA (CRS): 45,596 kg
- 3- Average payload for F9 v1.1: 2,928.4 kg
- 4- First successful ground pad landing: 2015-12-22
- 5- Maximum payload mass: 15,600 kg

GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map:

Object Type	What Was Added	Why It Was Added
Markers	Red markers at each launch site location	To visually identify exact positions of CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E
Color-coded Markers	Green markers for successful landings, Red markers for failed landings	To distinguish launch outcomes at a glance
Marker Cluster	Grouped multiple markers at same location	To prevent overlapping markers and improve map readability when multiple launches occurred at same site

GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Build an Interactive Map with Folium

Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map:

Object Type	What Was Added	Why It Was Added
Circles	100km and 500km radius circles around each launch site	To visualize proximity ranges and understand geographic constraints
Lines (PolyLine)	Lines connecting launch sites to coastline, railway, and highway points	To calculate and display exact distances to nearby infrastructure
Distance Labels	DivIcon markers showing distance in km	To display numerical distance values directly on the map
MousePosition	Tool showing latitude/longitude on mouse hover	To easily capture coordinates of any point on the map for distance calculations

GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Build an Interactive Map with Folium

Key Findings from Folium Map:

- All launch sites are within 3-12 km of coastline (launch over water for safety)
- Launch sites are near railways and highways (transportation of rocket components)
- KSC LC-39A shows highest concentration of green (successful) markers
- Florida sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) are within 6-7 km of each other

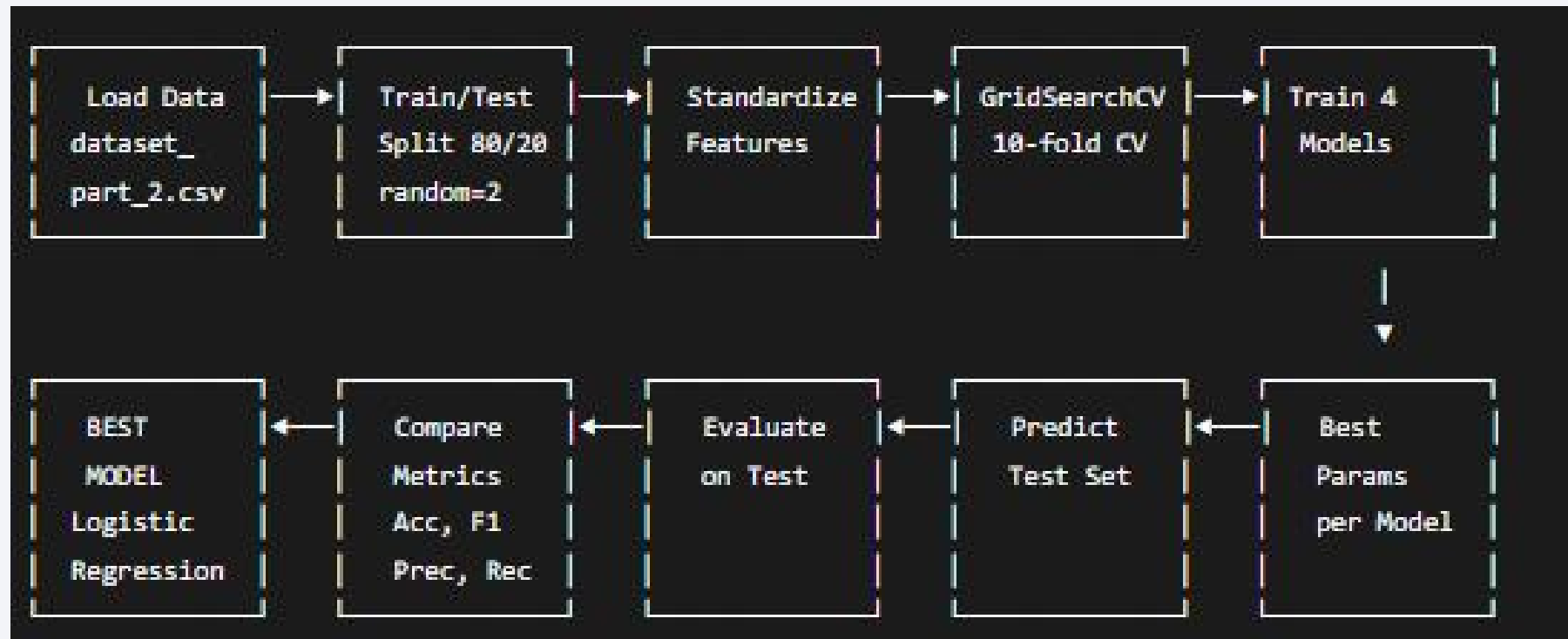
GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
- You need present your model development process using key phrases and flowchart
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

Predictive Analysis (Classification)



GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

Model Performance Summary:

Model	Best Parameters	Test Accuracy	F1 Score
Logistic Regression	C=0.01, l2, lbfgs	83.33%	0.8889
SVM	C=1, gamma=0.0316, sigmoid	83.33%	0.8889
Decision Tree	gini, depth=8, min_samples_leaf=4	77.78%	0.8462
KNN	n_neighbors=10, p=1	83.33%	0.8889

Best Model: Logistic Regression (83.33% test accuracy)

Confusion Matrix: $\begin{bmatrix} 3 & 3 \\ 0 & 12 \end{bmatrix}$ → 12 correct predictions, 3 false positives

GitHub URL

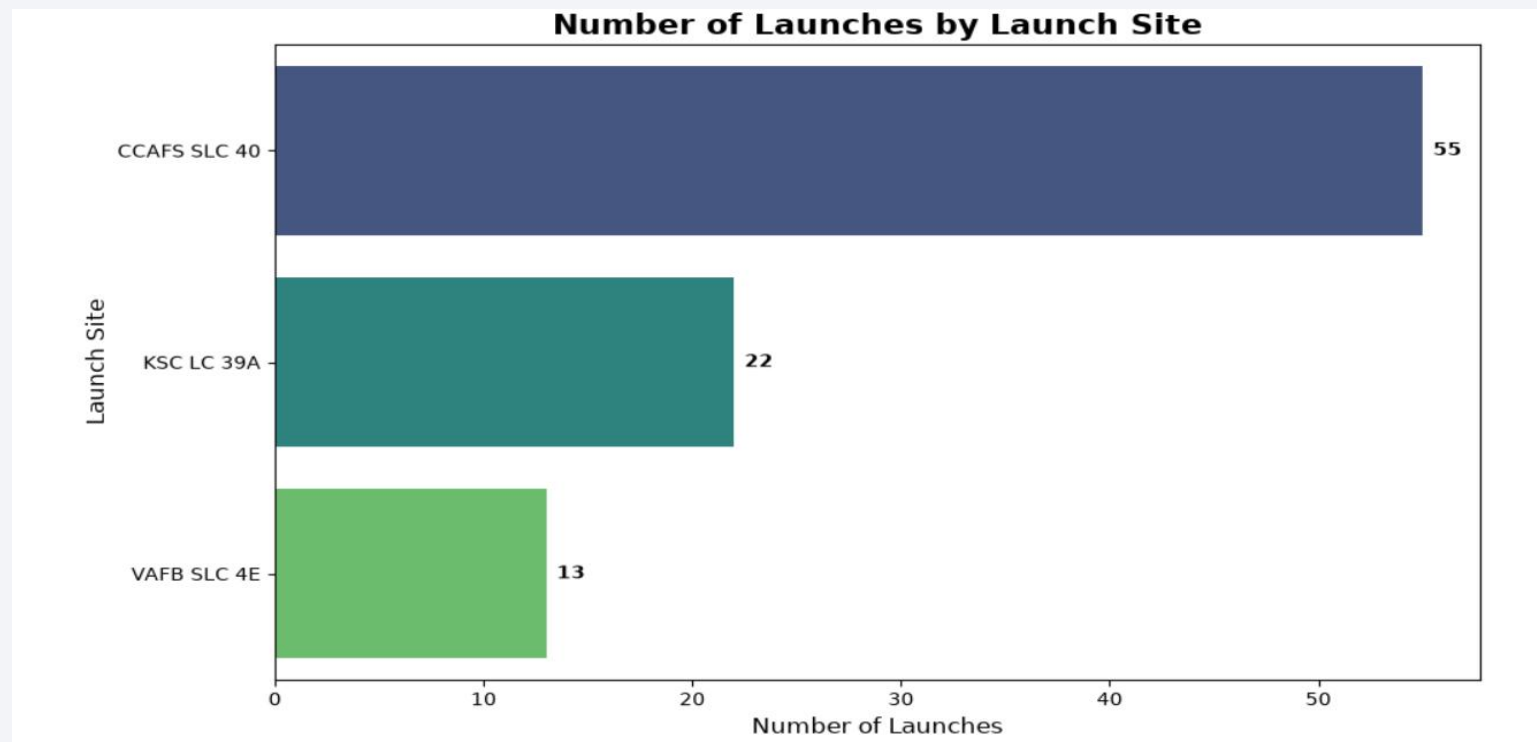
https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Results

Number of Launches Launch Site

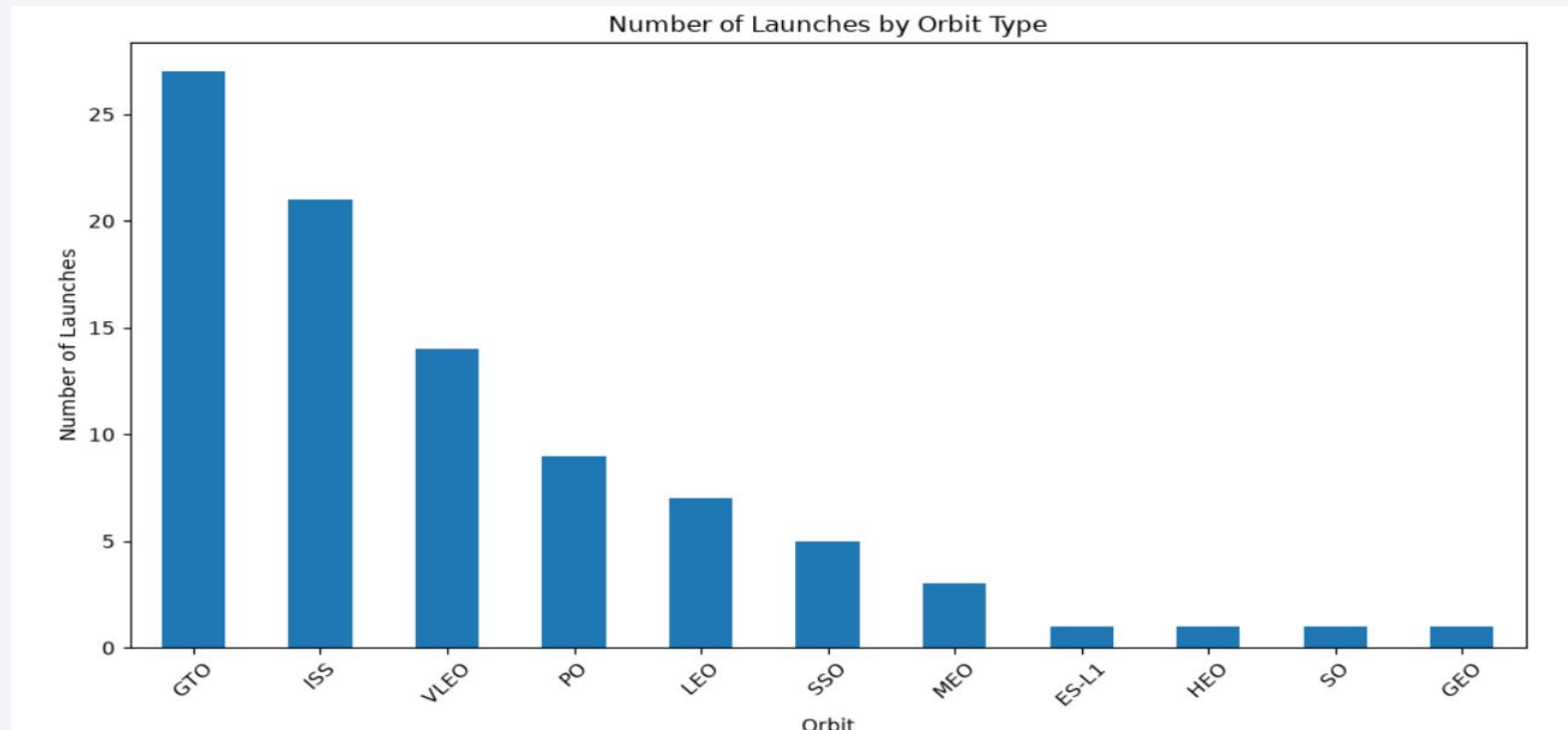


GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Results

Number of Launches by Orbit Type

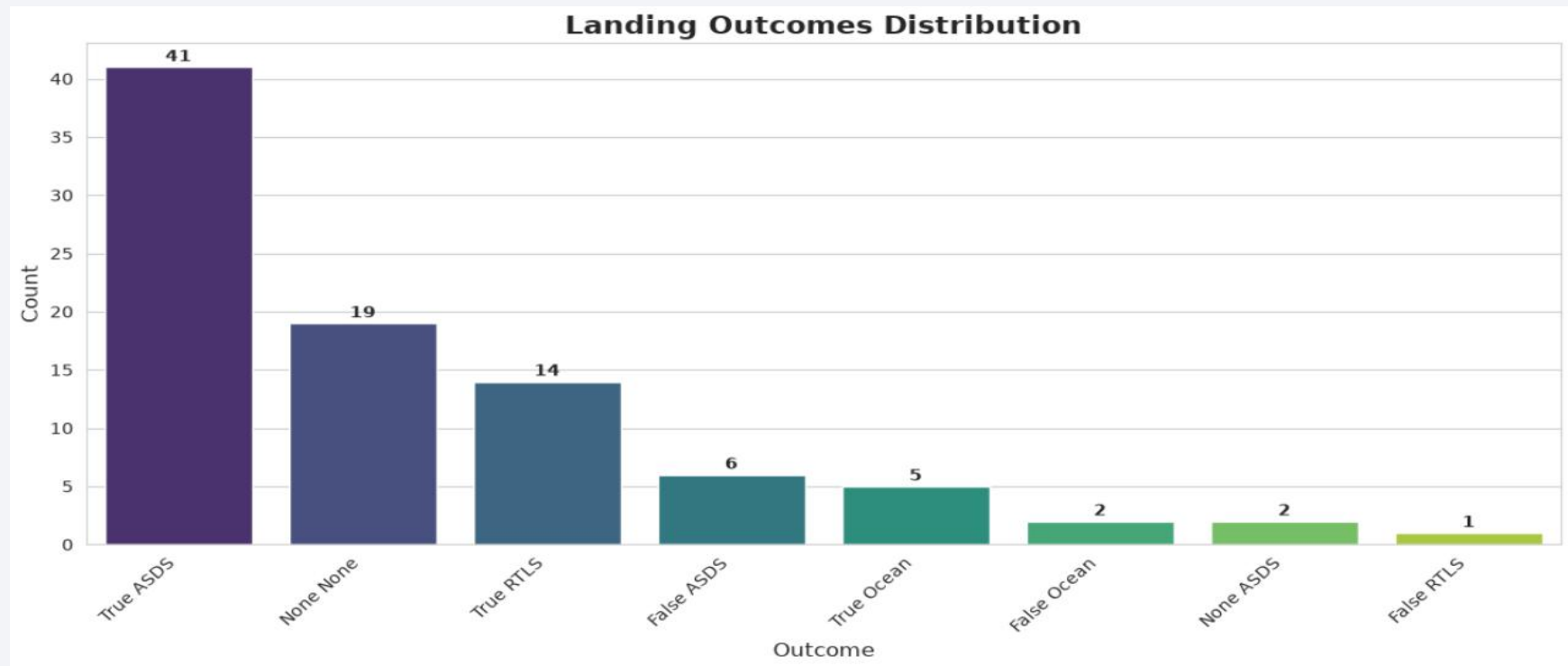


GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Results

Landing Outcomes Distribution

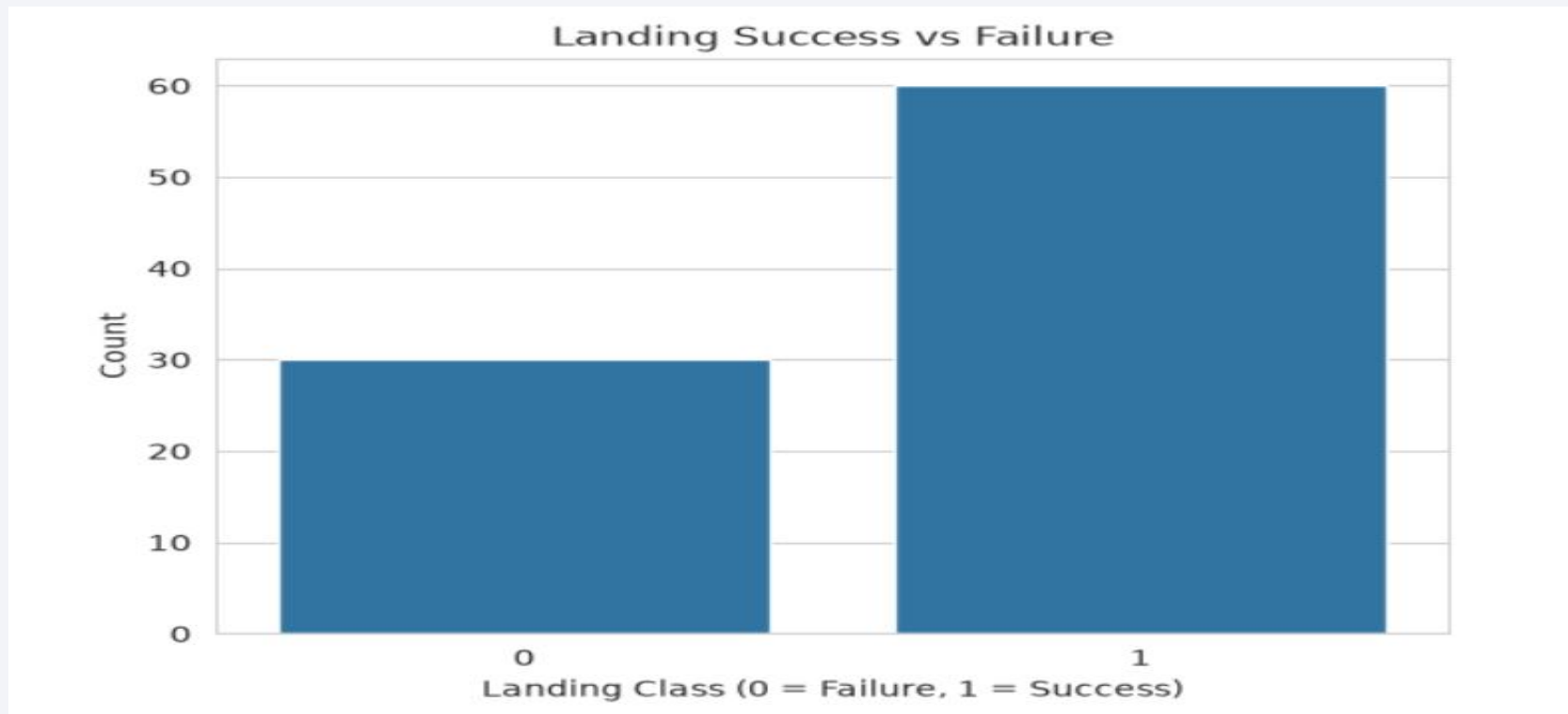


GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Results

Landing Success VS failure

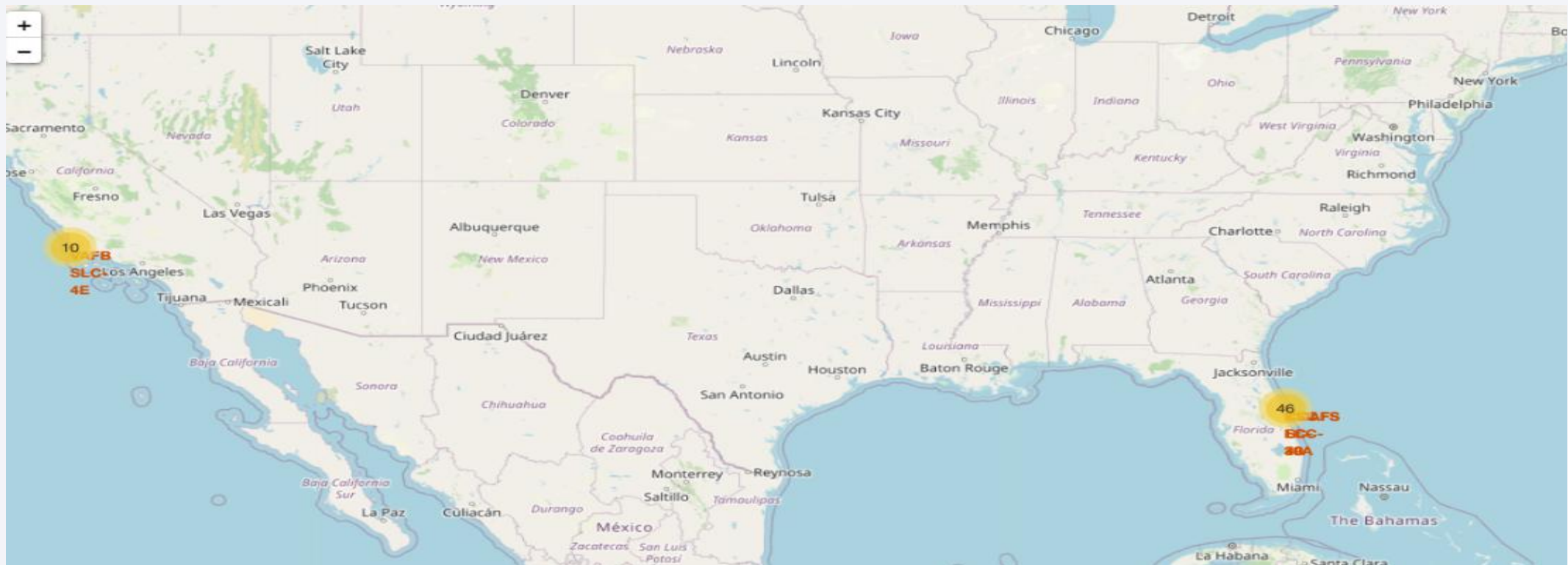


GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Results

Map1 :Launch sites across US (Florida + California)

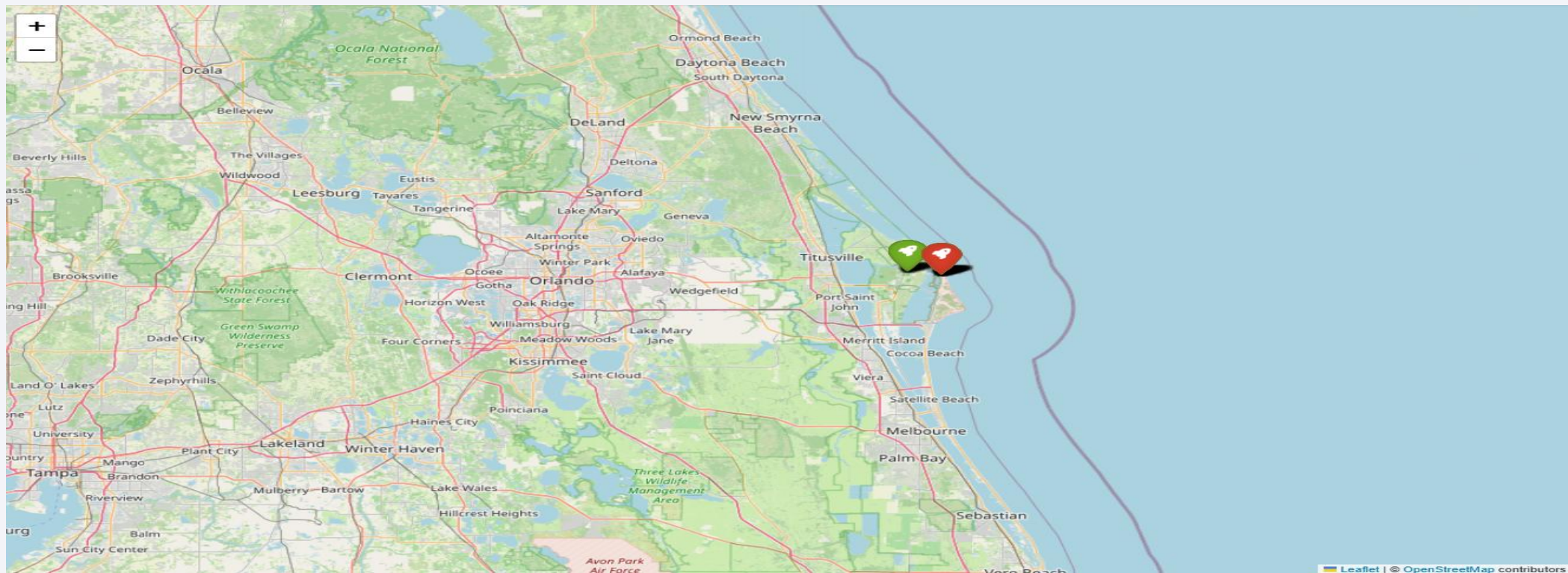


GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Results

Map2 :Color-coded success (green) vs failure (red)

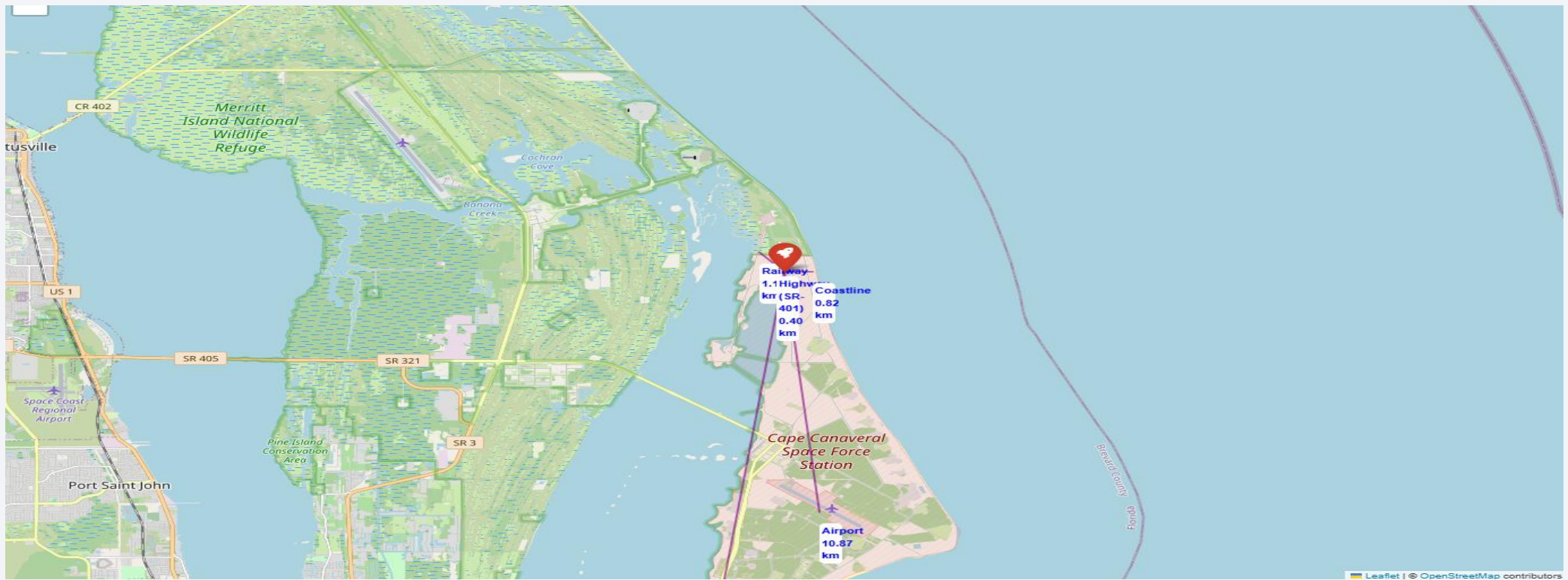


GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Results

Map3 :Launch site + Coastline (10.87 km) + Highway + Railway

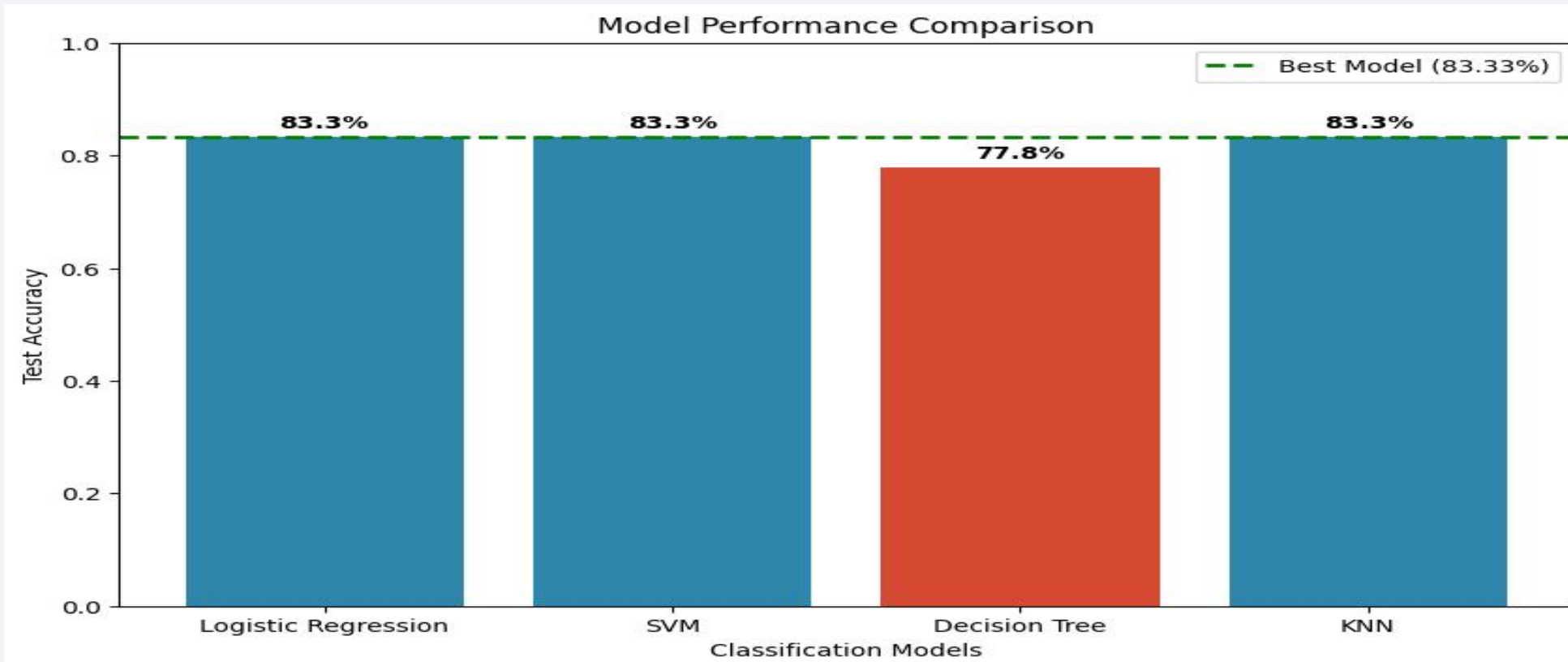


GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Results

Model Performance Comparison:

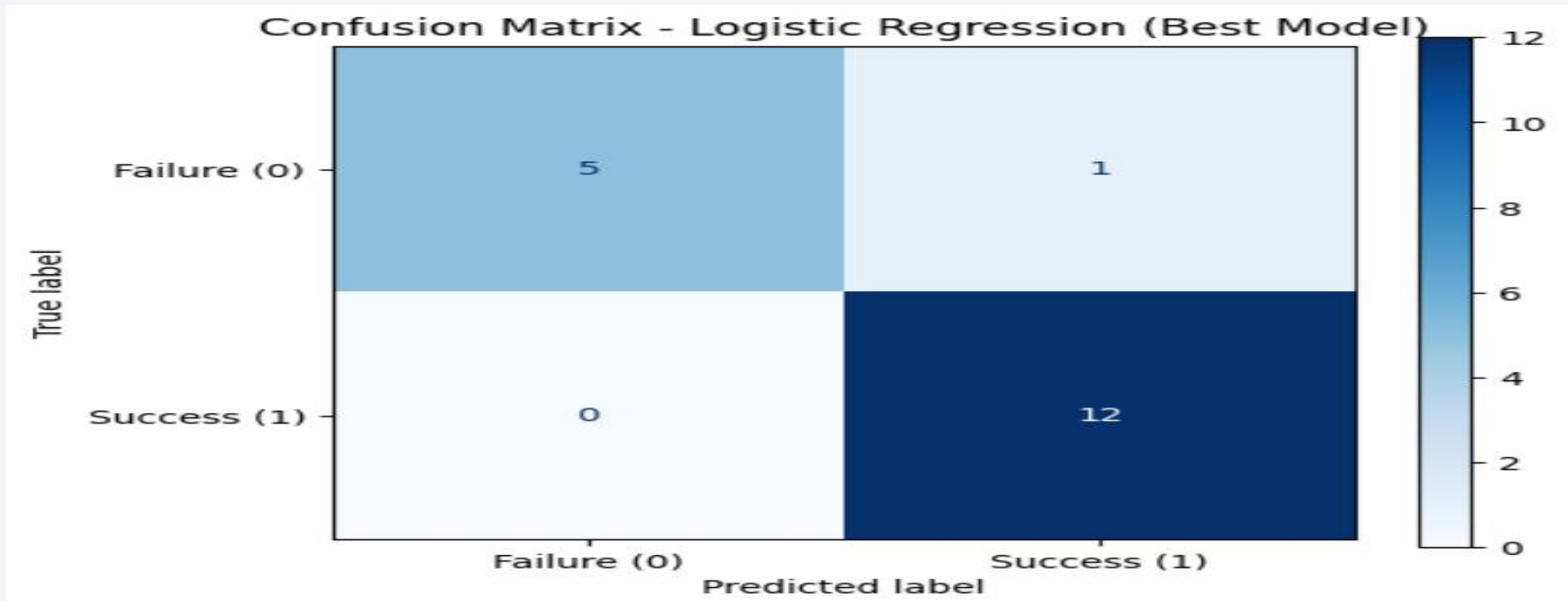


GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

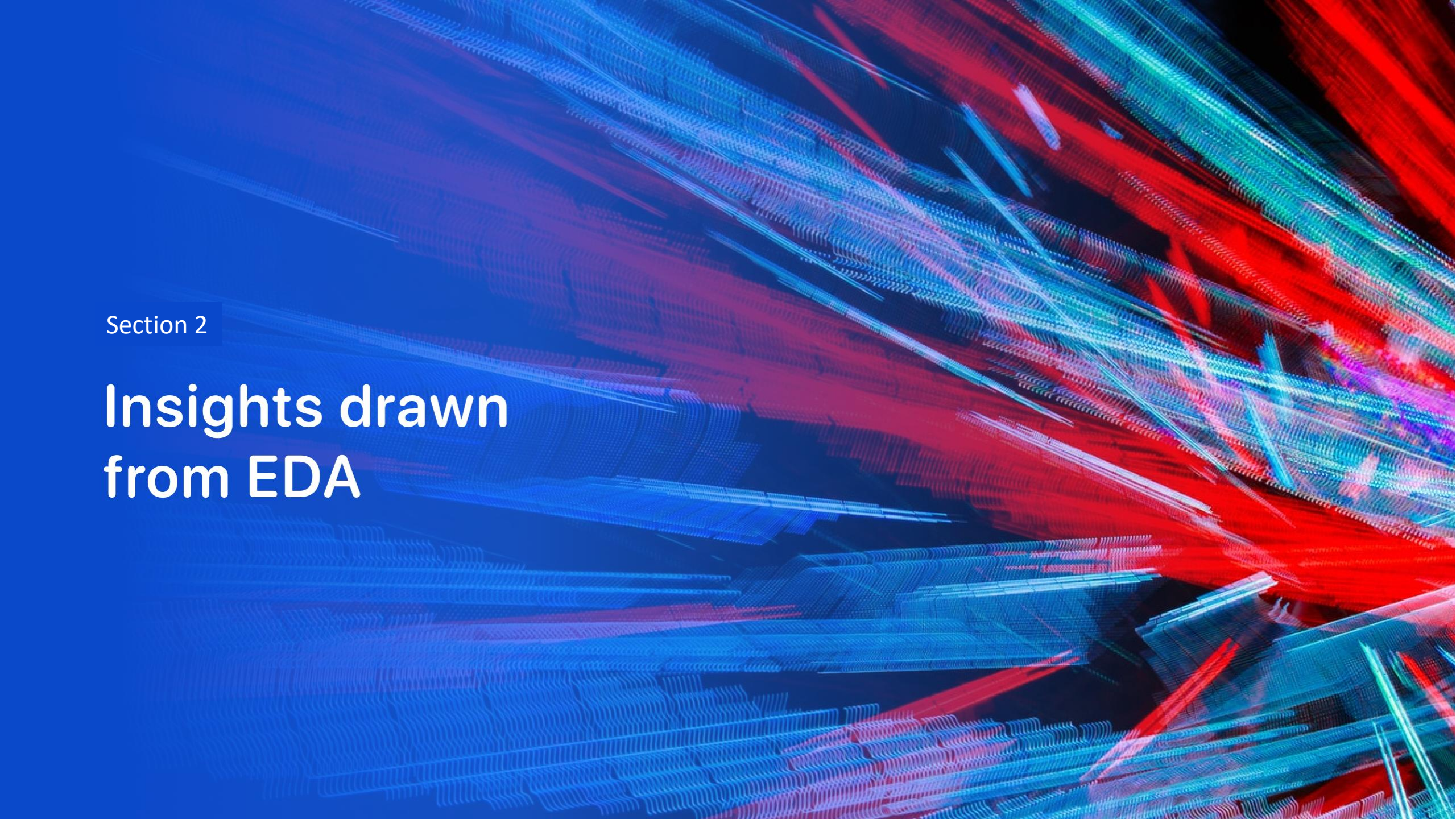
Results

Confusion Matrix - Logistic Regression (Best Model)



GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

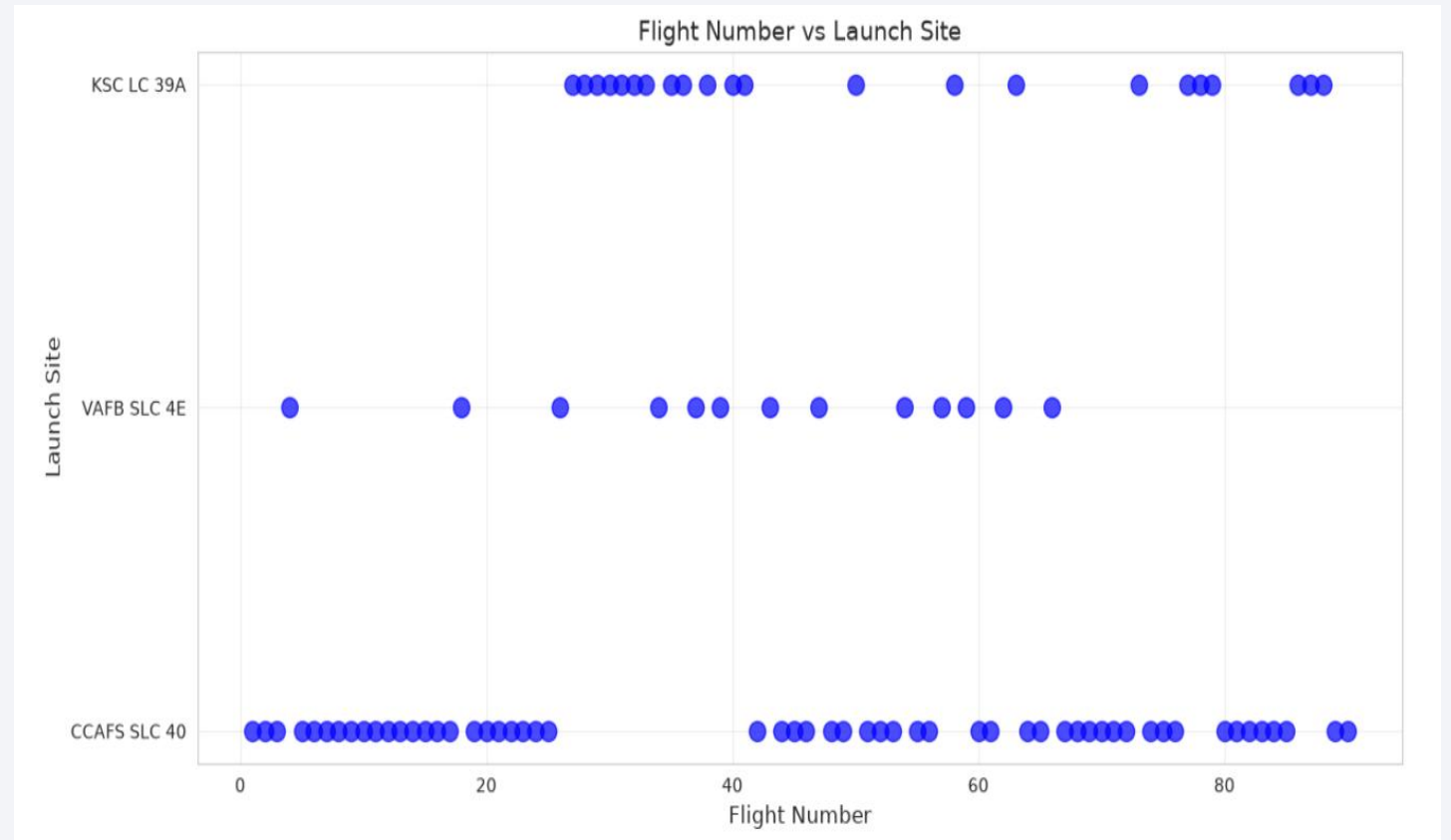
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

The scatter plot displays flight numbers on the X-axis and launch sites on the Y-axis. CCAFS SLC-40 was the most frequently used site, with launches across all flight numbers. KSC LC-39A began operations around flight 40, while VAFB SLC-4E appears less often, used primarily for polar orbit missions. This shows SpaceX's strategic distribution of launches across multiple sites.

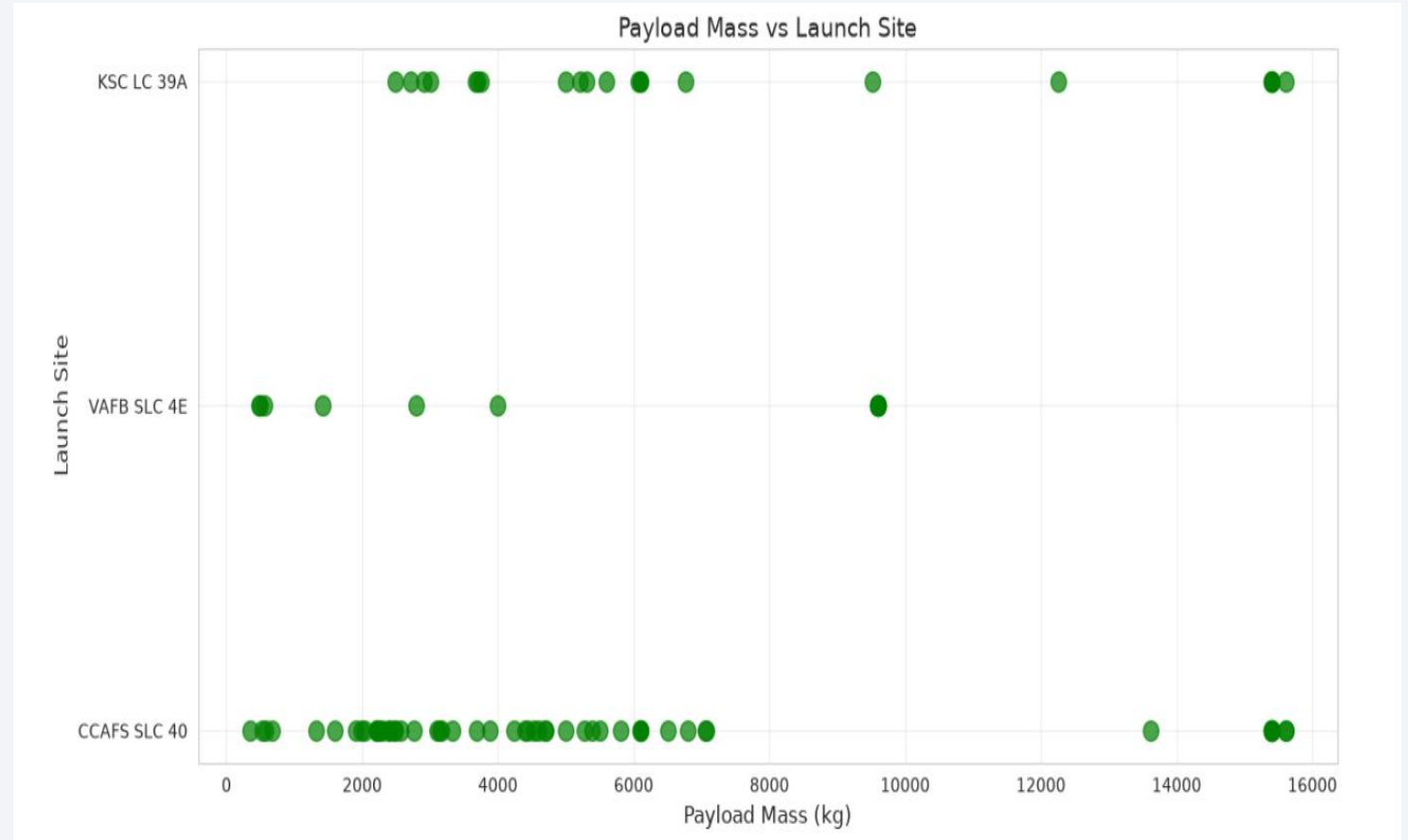


GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Payload vs. Launch Site

The scatter plot shows payload mass distribution across launch sites. KSC LC-39A handled the heaviest payloads (up to 15,600 kg), while VAFB SLC-4E only launched light payloads under 500 kg for polar orbits. CCAFS LC-40 and CCAFS SLC-40 covered a broad mid-range between 0 and 6,000 kg. This indicates that heavier payloads requiring more thrust are launched from Florida sites, while lighter polar missions use Vandenberg.

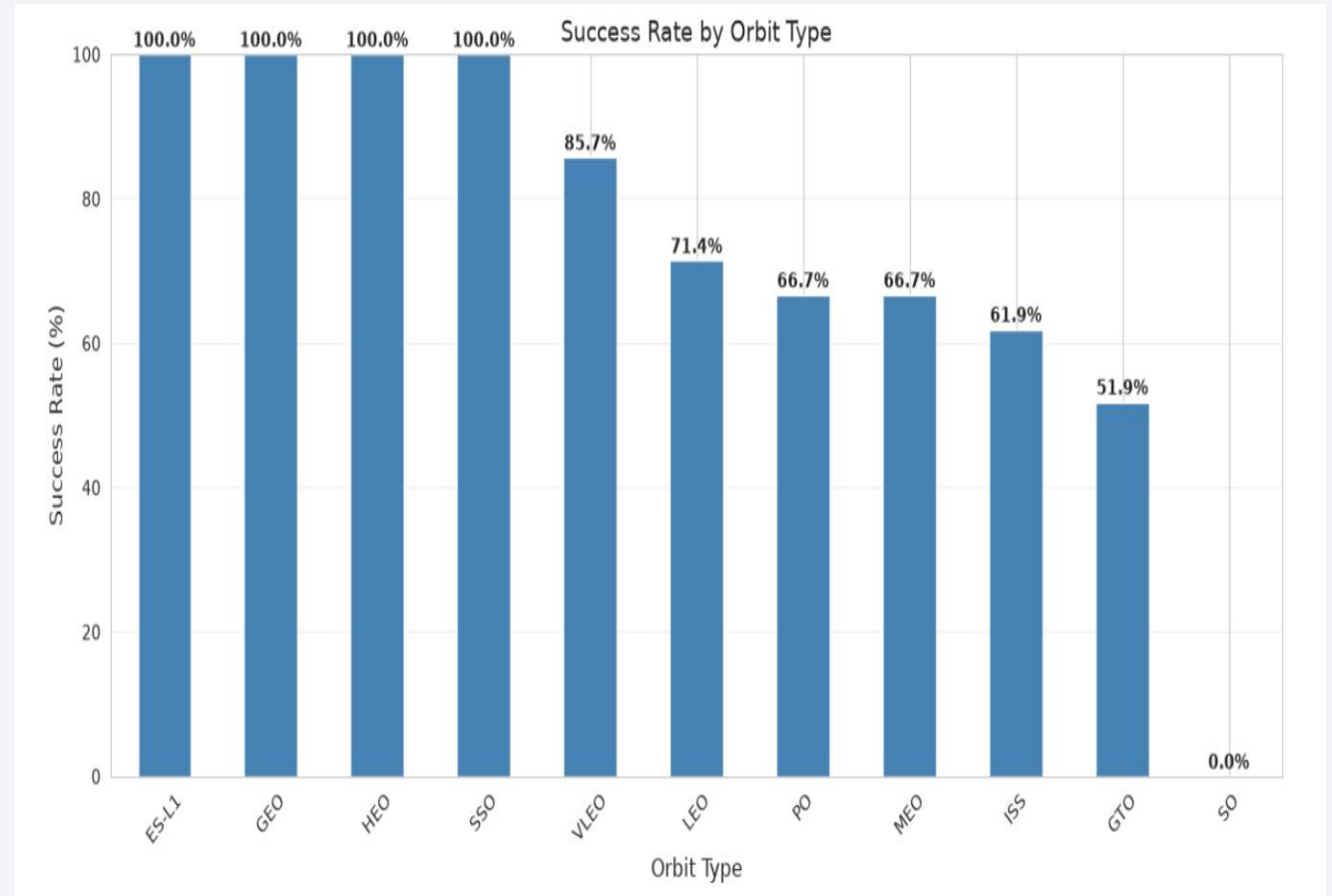


GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Success Rate vs. Orbit Type

The bar chart displays success rates for each orbit type. ES-L1, GEO, HEO, and SSO achieved 100% success rates. VLEO followed with 85.7%, while LEO had 71.4%. PO and MEO both achieved 66.7%, ISS had 61.9%, and GTO showed 51.9%. SO had 0% success rate. This indicates that specialized orbits like ES-L1 and GEO have higher success rates, while GTO missions are more challenging for booster landing due to higher energy requirements.

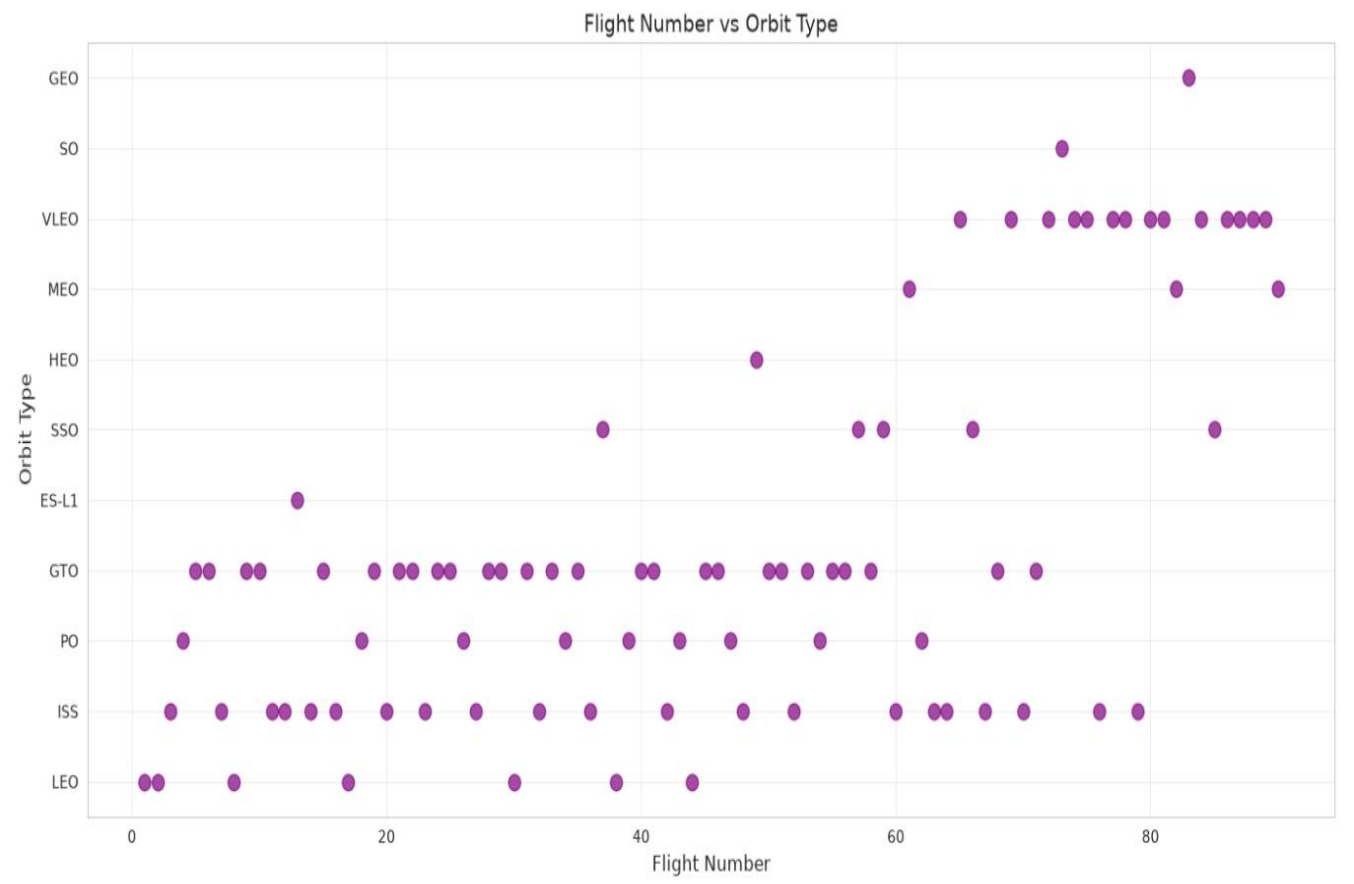


GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

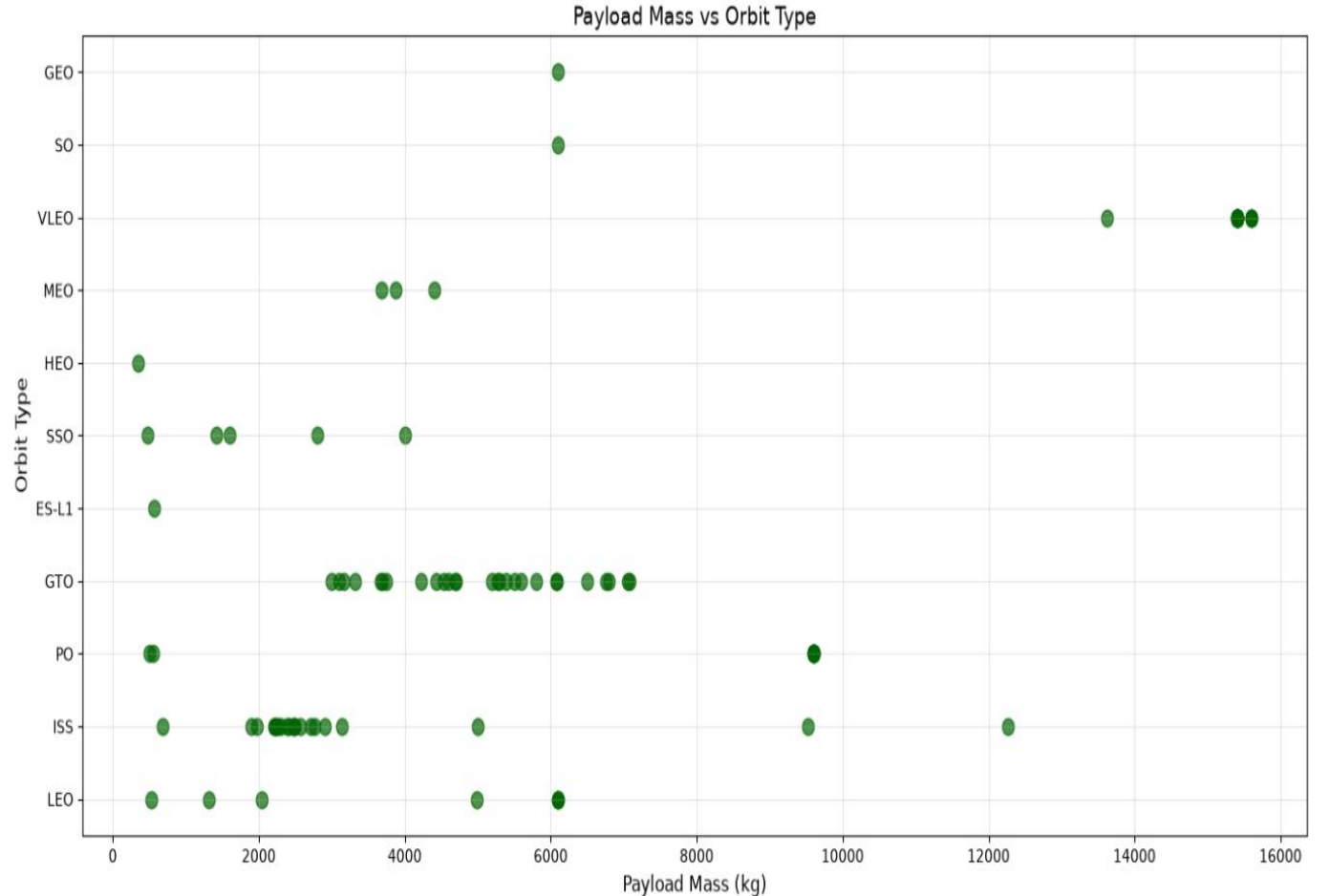
Flight Number vs. Orbit Type

The scatter plot displays orbit types across flight numbers using color-coded points. Early missions (flights 1-20) primarily targeted LEO and ISS orbits shown in blue and orange. GTO missions, shown in green, appeared consistently throughout the program. Specialized orbits such as ES-L1, HEO, MEO, GEO, and SO occurred less frequently, representing unique customer requirements. The increasing diversity of orbit types over time demonstrates SpaceX's growing capability to serve various mission profiles, from low Earth orbit satellites to deep space and polar operations.



Payload vs. Orbit Type

The scatter plot shows payload mass distribution across orbit types. GTO missions carry payloads ranging from 1,000 to 6,000 kg, requiring significant thrust. LEO and ISS orbits accommodate the widest range, from 0 to 15,600 kg, demonstrating versatility for various customer needs. Polar and sun-synchronous orbits (PO, SSO, VLEO) typically carry lighter payloads under 1,000 kg. Specialized orbits like ES-L1, HEO, and GEO fall in the medium range of 500 to 2,000 kg. This indicates that heavier payloads generally target lower Earth orbits, while lighter payloads are sufficient for higher or specialized orbits.

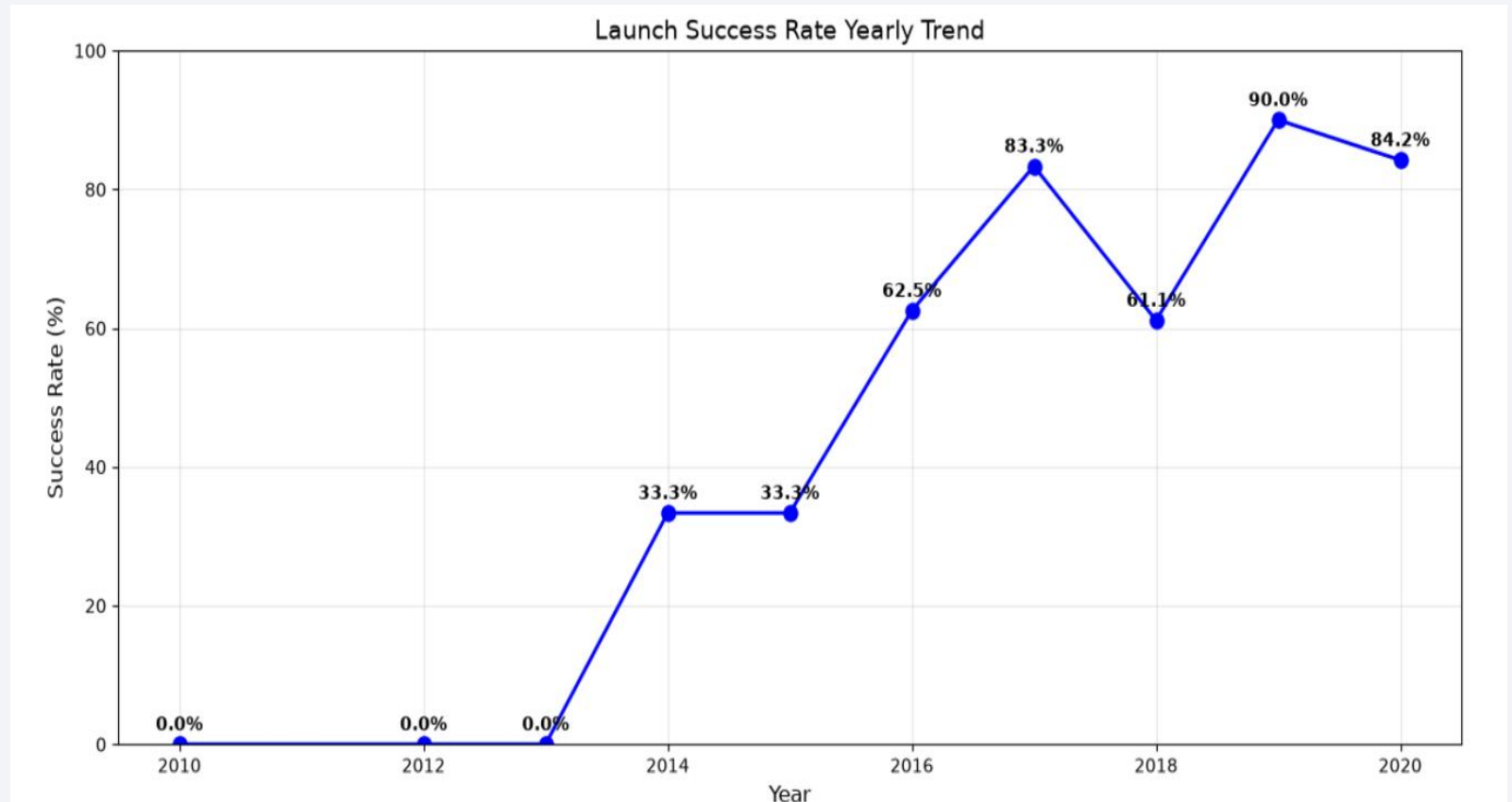


GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Launch Success Yearly Trend

The line chart shows the yearly average success rate of Falcon 9 first stage landings from 2010 to 2020. Early years (2010-2013) had 0% success due to experimental launches. Success rates improved significantly after 2014, reaching 100% in 2016, 2017, and 2019. The overall trend demonstrates a clear upward trajectory, with success rates stabilizing above 80% after 2015.



GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

All Launch Site Names

The query returns four unique launch sites used by SpaceX: two at Cape Canaveral (CCAFS LC-40 and CCAFS SLC-40), one at Kennedy Space Center (KSC LC-39A) in Florida, and one at Vandenberg Air Force Base (VAFB SLC-4E) in California.

```
import pandas as pd
import sqlite3

# Connect to database
conn = sqlite3.connect('spacex.db')

# Task 1: Unique launch sites
result = pd.read_sql_query("SELECT DISTINCT Launch_Site FROM SPACEXTABLE;", conn)
print(result)
```

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Launch Site Names Begin with 'CCA'

The query returns five records where launch sites begin with "CCA", all from CCAFS LC-40. The results show early Falcon 9 launches between 2010 and 2013, all using F9 v1.0 booster versions B0003 through B0007.

```
# Task 2: Display 5 records where launch sites begin with 'CCA'
result = pd.read_sql_query("SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;", conn)
print("Task 2 - 5 records with launch sites starting with 'CCA':")
print(result)
```

Task 2 - 5 records with launch sites starting with 'CCA':

	Date	Time (UTC)	Booster_Version	Launch_Site	\
0	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	
1	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	
2	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	
3	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	
4	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	

Total Payload Mass

The query displays the column structure of the SPACEXTABLE, showing 10 columns including Date, Time, Booster_Version, Launch_Site, Payload, PAYLOAD_MASS_KG, Orbit, Customer, Mission_Outcome, and Landing_Outcome. The PAYLOAD_MASS_KG column is an integer type containing the payload mass in kilograms for each mission.

```
# Check column names in the table
result = pd.read_sql_query("PRAGMA table_info(SPACEXTABLE);", conn)
print(result)
```

	cid	name	type	notnull	dflt_value	pk
0	0	Date	TEXT	0	None	0
1	1	Time (UTC)	TEXT	0	None	0
2	2	Booster_Version	TEXT	0	None	0
3	3	Launch_Site	TEXT	0	None	0
4	4	Payload	TEXT	0	None	0
5	5	PAYLOAD_MASS_KG_	INTEGER	0	None	0
6	6	Orbit	TEXT	0	None	0
7	7	Customer	TEXT	0	None	0
8	8	Mission_Outcome	TEXT	0	None	0
9	9	Landing_Outcome	TEXT	0	None	0

GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Average Payload Mass by F9 v1.1

```
# Task 4: Average payload mass for F9 v1.1
result = pd.read_sql_query("SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';", conn)
print("Task 4 - Average payload mass for F9 v1.1:")
print(result)
```

```
Task 4 - Average payload mass for F9 v1.1:
  AVG(PAYLOAD_MASS_KG_)
0                2928.4
```

The query calculates the average payload mass carried by the F9 v1.1 booster version. The result shows that F9 v1.1 boosters carried an average payload mass of 2,928.4 kg across their missions.

First Successful Ground Landing Date

```
result = pd.read_sql_query("SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';", conn)
print("Task 5 - First successful ground pad landing:")
print(result)
```

```
Task 5 - First successful ground pad landing:
      MIN(Date)
0  2015-12-22
```

The query finds the earliest date when a successful ground pad landing occurred. The result shows that the first successful ground pad landing was achieved on December 22, 2015, marking a significant milestone in SpaceX's reusability program.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
result = pd.read_sql_query("SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG BETWEEN 4000 AND 6000;", conn)
print("Task 6 - Boosters with success on drone ship and payload 4000-6000 kg:")
print(result)
```

Task 6 - Boosters with success on drone ship and payload 4000-6000 kg:

	Booster_Version
0	F9 FT B1022
1	F9 FT B1026
2	F9 FT B1021.2
3	F9 FT B1031.2

The query lists booster versions that successfully landed on a drone ship while carrying payloads between 4,000 and 6,000 kg. The results show four FT boosters: B1022, B1026, B1021.2, and B1031.2, demonstrating the reliability of the FT version for medium-to-heavy payload missions.

Total Number of Successful and Failure Mission Outcomes

```
result = pd.read_sql_query("SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE GROUP BY Mission_Outcome;", conn)
print("Task 7 - Mission outcomes count:")
print(result)
```

Task 7 - Mission outcomes count:

	Mission_Outcome	COUNT(*)
0	Failure (in flight)	1
1	Success	98
2	Success	1
3	Success (payload status unclear)	1

The query counts the number of missions for each outcome type. Results show 98 successful missions, 1 failure during flight, 1 additional success entry, and 1 success with unclear payload status, demonstrating SpaceX's high overall mission success rate.

GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Boosters Carried Maximum Payload

```
Task 8 - Booster versions with maximum payload mass:
Booster_Version  PAYLOAD_MASS__KG_
0      F9 B5 B1048.4      15600
1      F9 B5 B1049.4      15600
2      F9 B5 B1051.3      15600
3      F9 B5 B1056.4      15600
4      F9 B5 B1048.5      15600
5      F9 B5 B1051.4      15600
6      F9 B5 B1049.5      15600
7      F9 B5 B1060.2      15600
8      F9 B5 B1058.3      15600
9      F9 B5 B1051.6      15600
10     F9 B5 B1060.3      15600
11     F9 B5 B1049.7      15600
```

The query identifies booster versions that carried the maximum payload mass of 15,600 kg. All are F9 B5 boosters including versions B1048, B1049, B1051, B1056, B1058, and B1060, demonstrating that the B5 block is SpaceX's most powerful and capable booster for heavy payload missions.

2015 Launch Records

```
result = pd.read_sql_query("SELECT substr(Date, 6, 2) as month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date, 1, 4) = '2015' AND Landin  
print("Task 9 - 2015 failures on drone ship:")  
print(result)
```

Task 9 - 2015 failures on drone ship:

	month	Landing_Outcome	Booster_Version	Launch_Site
0	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
1	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

The query extracts failed drone ship landings from 2015, showing the month, landing outcome, booster version, and launch site. Two failures occurred: one in January with booster B1012 and one in April with booster B1015, both at CCAFS LC-40 using F9 v1.1 boosters.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
result = pd.read_sql_query("SELECT Landing_Outcome, COUNT(*) as count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY count DESC")
print("Task 10 - Landing outcomes ranked:")
print(result)
```

Task 10 - Landing outcomes ranked:

	Landing_Outcome	count
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

The query ranks landing outcomes between June 4, 2010 and March 20, 2017. "No attempt" occurred most frequently (10 times), followed by "Success (drone ship)" and "Failure (drone ship)" with 5 each. "Success (ground pad)" and "Controlled (ocean)" had 3 each, while other outcomes like parachute failures and precluded landings had 1-2 occurrences. This shows that drone ship landings were the most common attempted recovery method during this period.

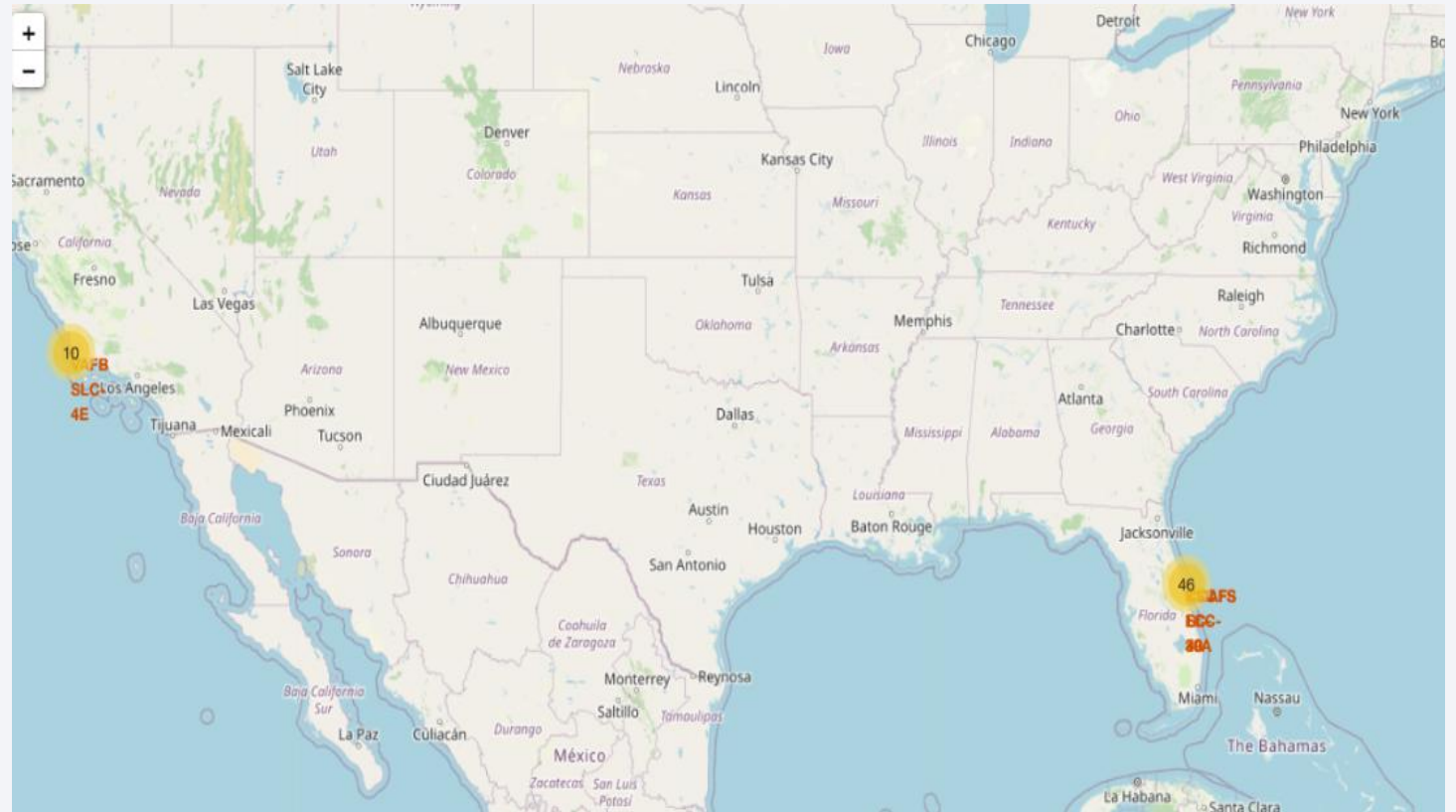
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a deep blue, with the horizon line visible. The city lights are concentrated in the lower right quadrant, showing a dense network of urban areas. The text "Section 3" is overlaid on the left side of the image.

Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

The map displays four SpaceX launch sites across the United States. Three sites are located on Florida's east coast - CCAFS LC-40, CCAFS SLC-40, and KSC LC-39A. One site, VAFB SLC-4E, is located on California's west coast. All launch sites are positioned near coastlines, allowing rockets to launch over water for safety and debris containment in case of failure.



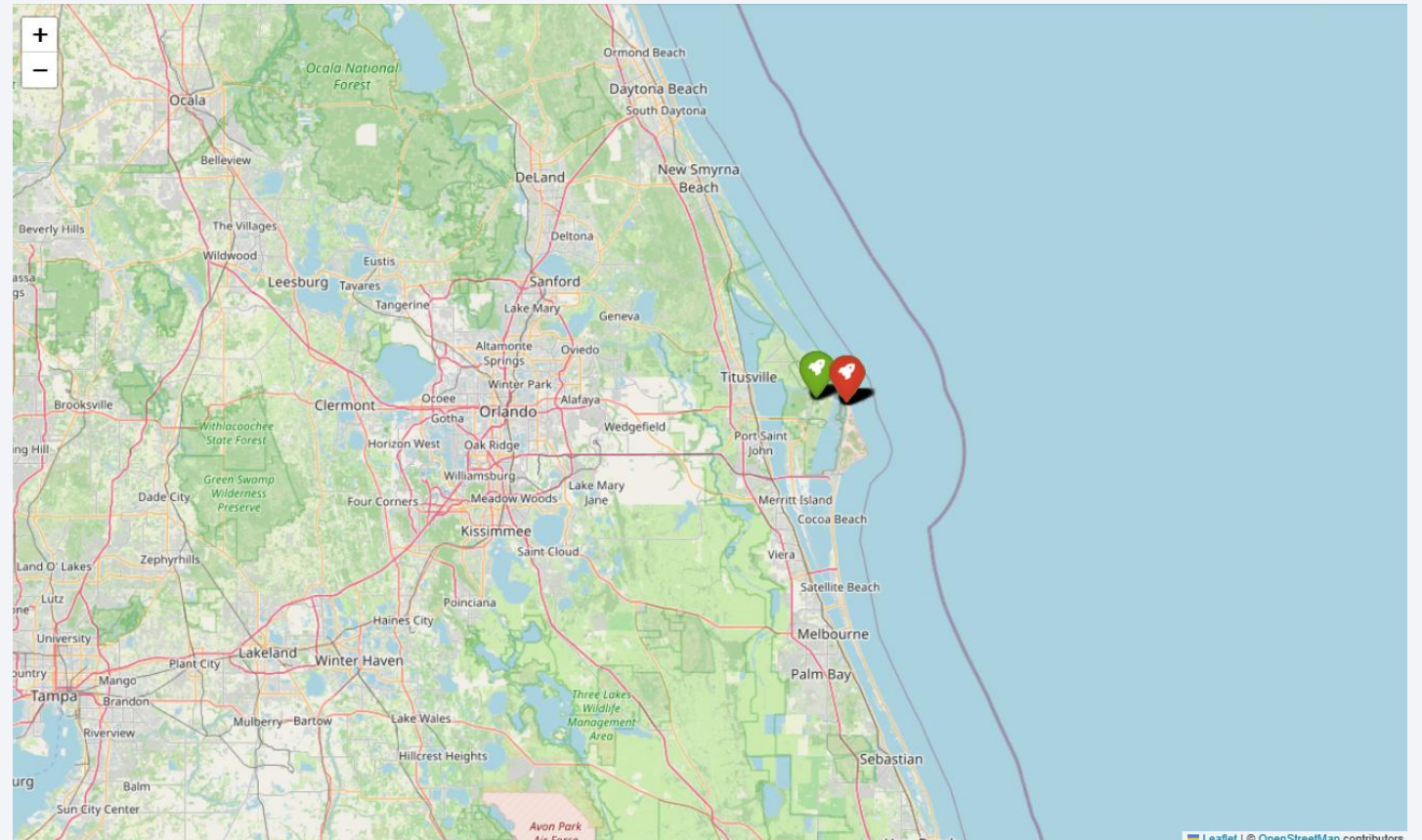
GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Results

Map2 :Color-coded success (green) vs failure (red)

The map displays green markers for successful landings and red markers for failed landings across all launch sites. Florida sites, particularly KSC LC-39A, show a higher concentration of green markers, indicating better success rates. California's VAFB SLC-4E shows a mix of both colors. The color-coding allows quick visual identification of which sites and which launches achieved successful booster recovery. Overall, the map demonstrates that success rates improved over time, with later launches showing more green markers.

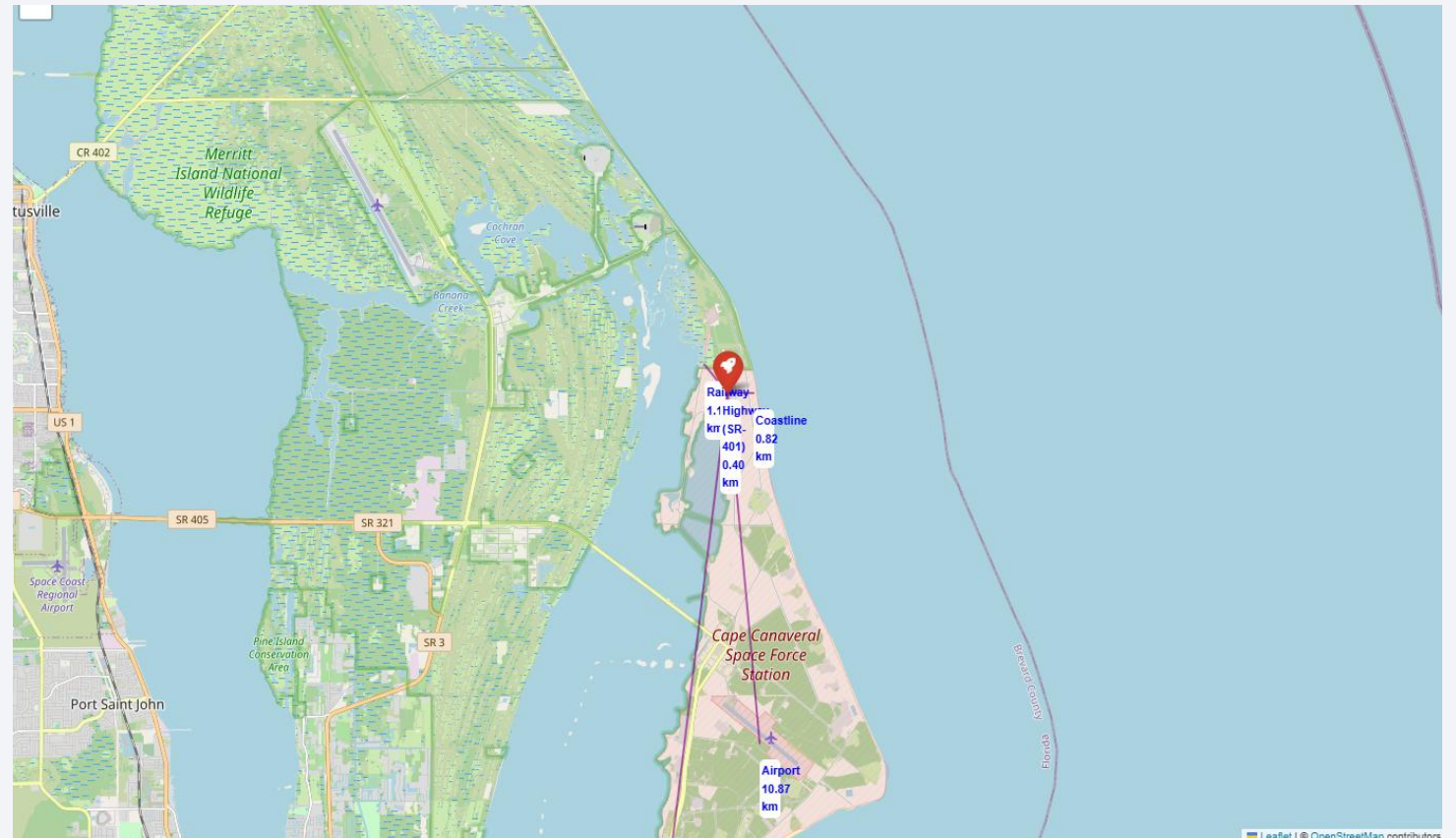


GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

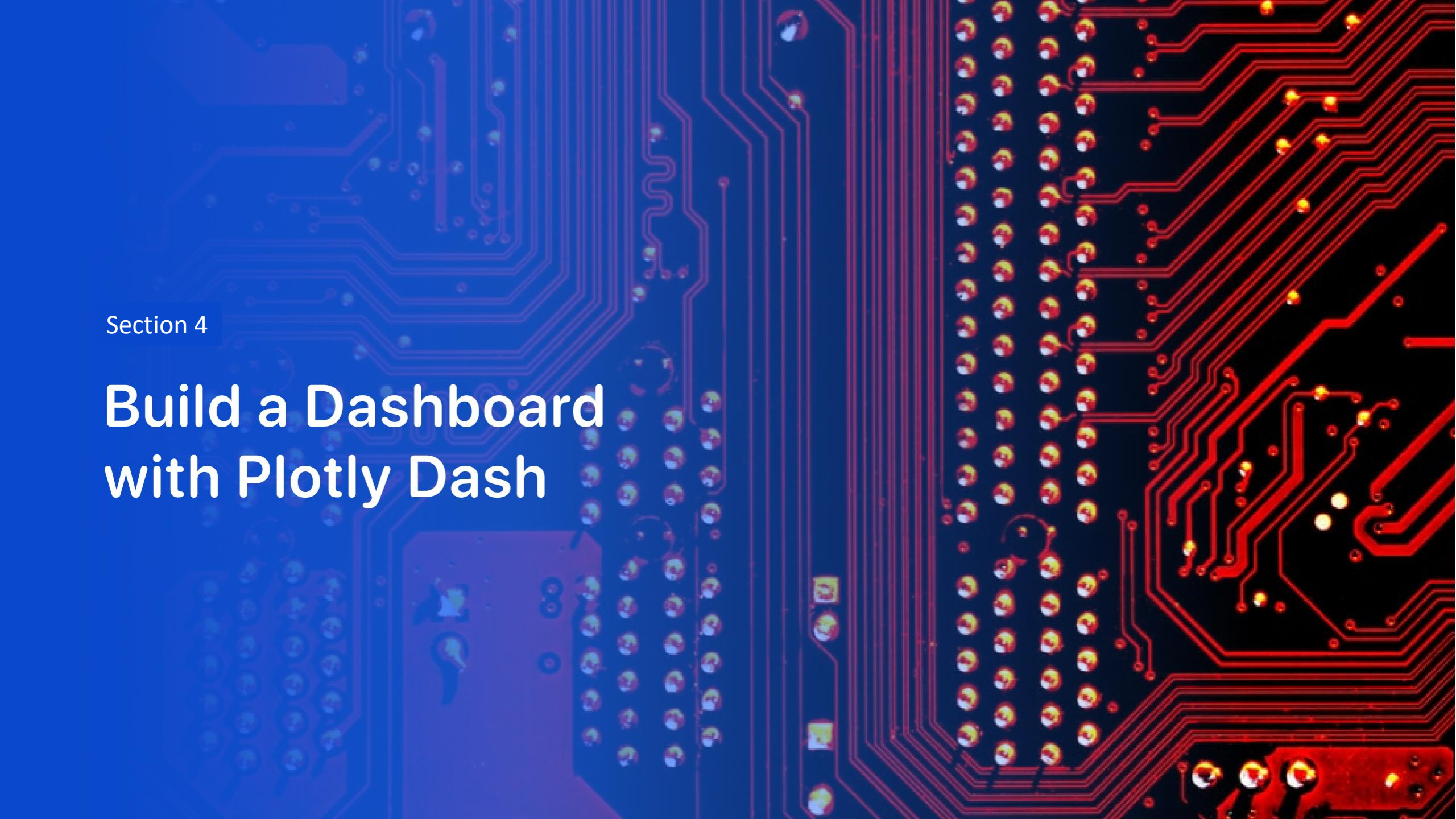
<Folium Map Screenshot 3>

The map displays CCAFS LC-40 launch site with calculated distances to nearby infrastructure. The coastline is approximately 10.87 km away. Highways SR405 and SR321 are within 0.40 to 0.52 km. A railway is approximately 1.1 km away. These short distances demonstrate that launch sites are strategically placed near transportation networks for delivering rocket components and near coastlines for launching over water, ensuring safety in case of failure.



GitHub URL

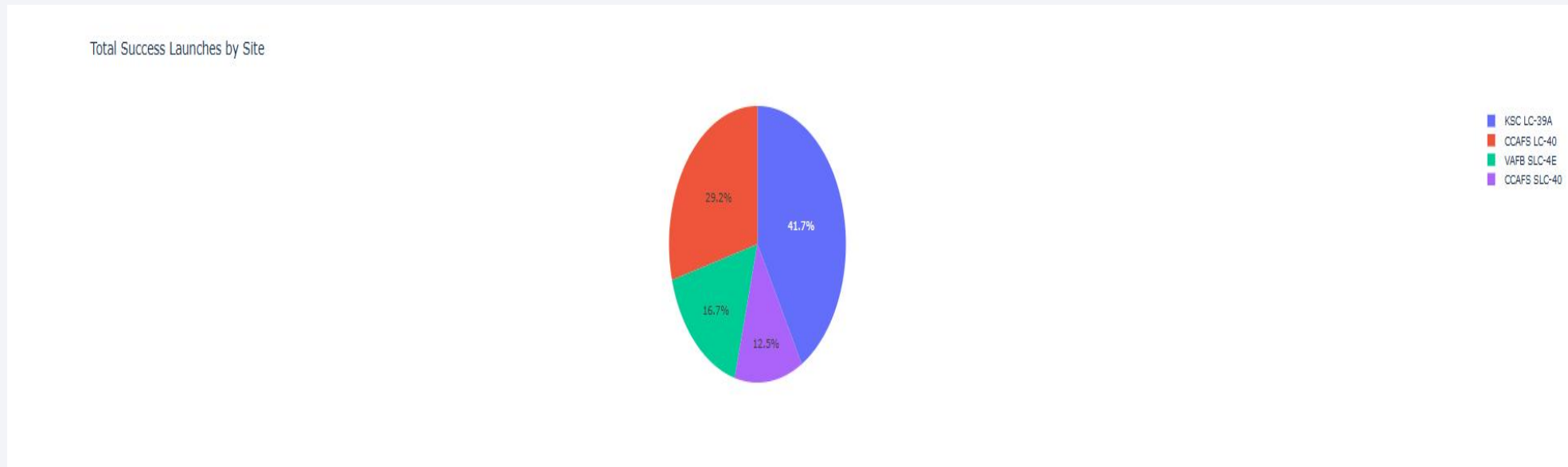
https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb



Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>



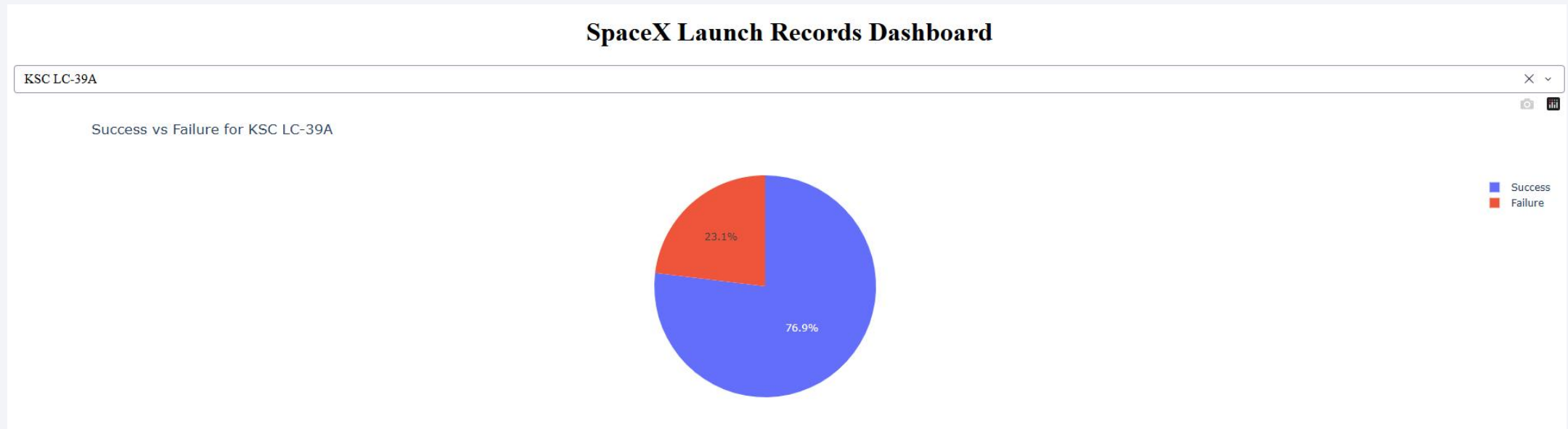
The pie chart displays the distribution of successful launches across four sites. KSC LC-39A leads with 41.7%, followed by CCAFS SLC-40 at 29.2%, VAFB SLC-4E at 16.7%, and CCAFS LC-40 at 12.5%. KSC LC-39A accounts for nearly half of all successful missions, indicating it is the most reliable launch site.

GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/spacex-dash-app.py>

<Dashboard Screenshot 2>

1.Success vs Failure for KSC LC-39A:



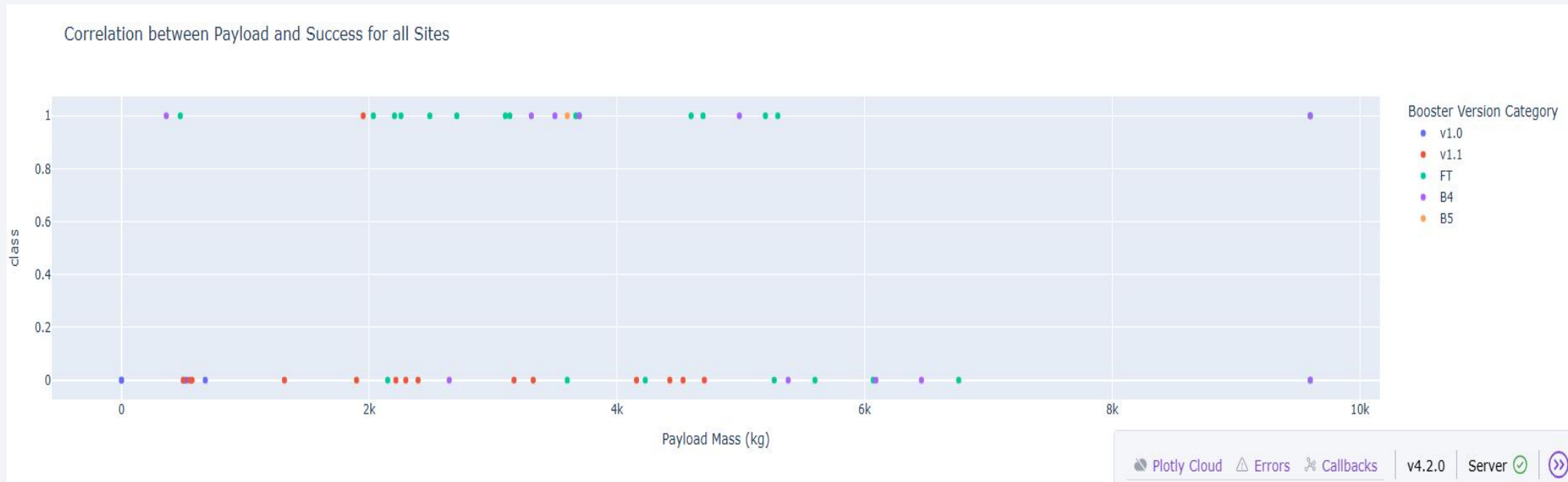
The pie chart shows success vs failure for KSC LC-39A, the launch site with the highest success ratio. Success accounts for 76.9% of launches, while failure accounts for only 23.1%. This indicates KSC LC-39A is the most reliable launch site, likely due to its newer infrastructure, experienced crew, and strategic location.

GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/spacex-dash-app.py>

<Dashboard Screenshot 3>

Show screenshots of Payload vs. Launch Outcome scatter plot for all site:



The scatter plot shows success (Class=1) concentrated in the 2,000-4,000 kg range, especially for FT and B5 boosters. FT boosters have the highest success rate with almost all points at Class=1. Light payloads (0-1,000 kg) using v1.0/v1.1 boosters show many failures. Mid-range payloads with advanced boosters have the highest success probability.

GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/spacex-dash-app.py>

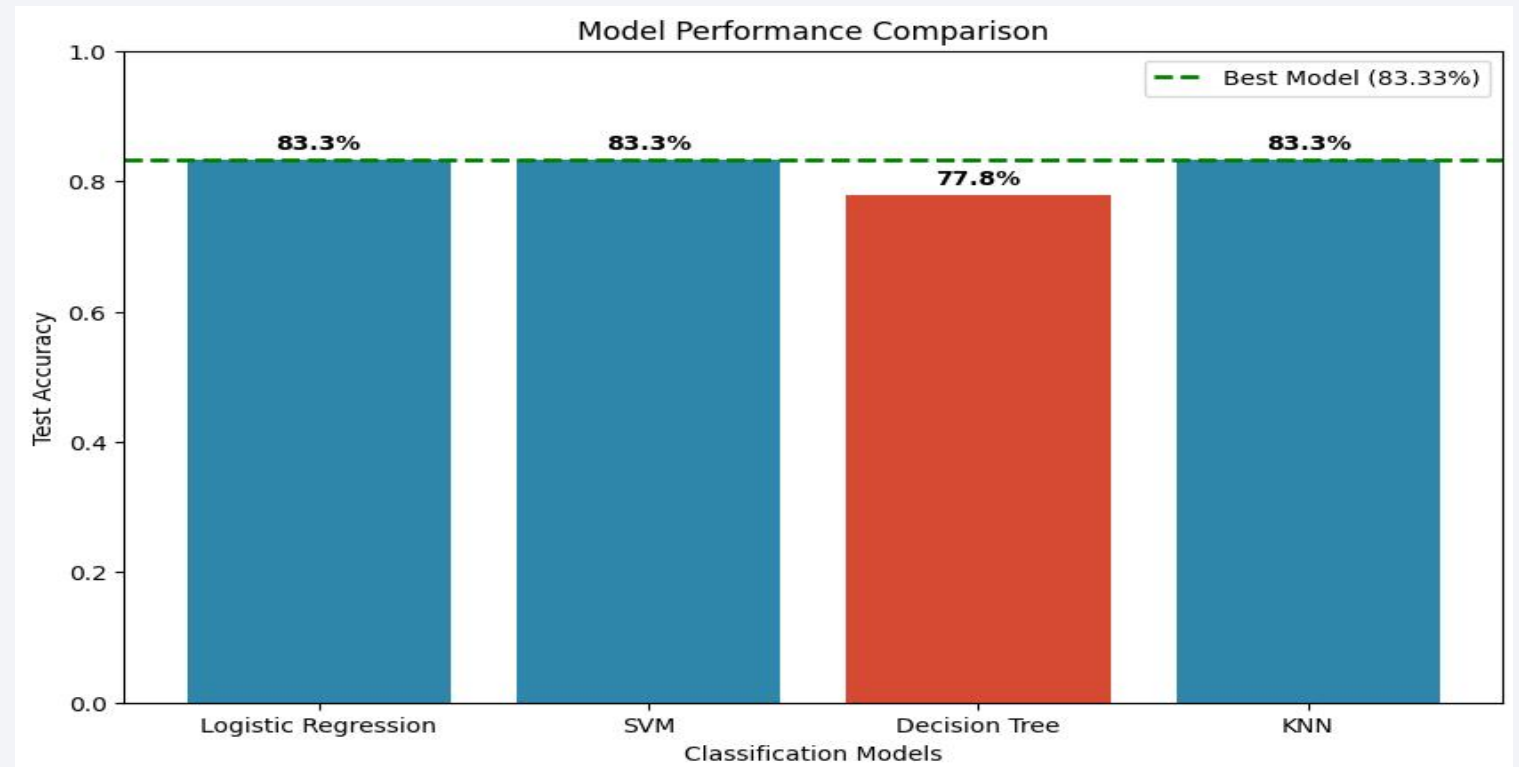
Section 5

Predictive Analysis (Classification)

Classification Accuracy

Model Performance Comparison on Test Data:

Three models (Logistic Regression, SVM, and KNN) achieved the same highest accuracy of 83.3%. Decision Tree performed slightly lower at 77.8%. Since multiple models achieved the same top accuracy, Logistic Regression is often preferred as the best model due to its simplicity, faster training time, and better interpretability compared to SVM and KNN.



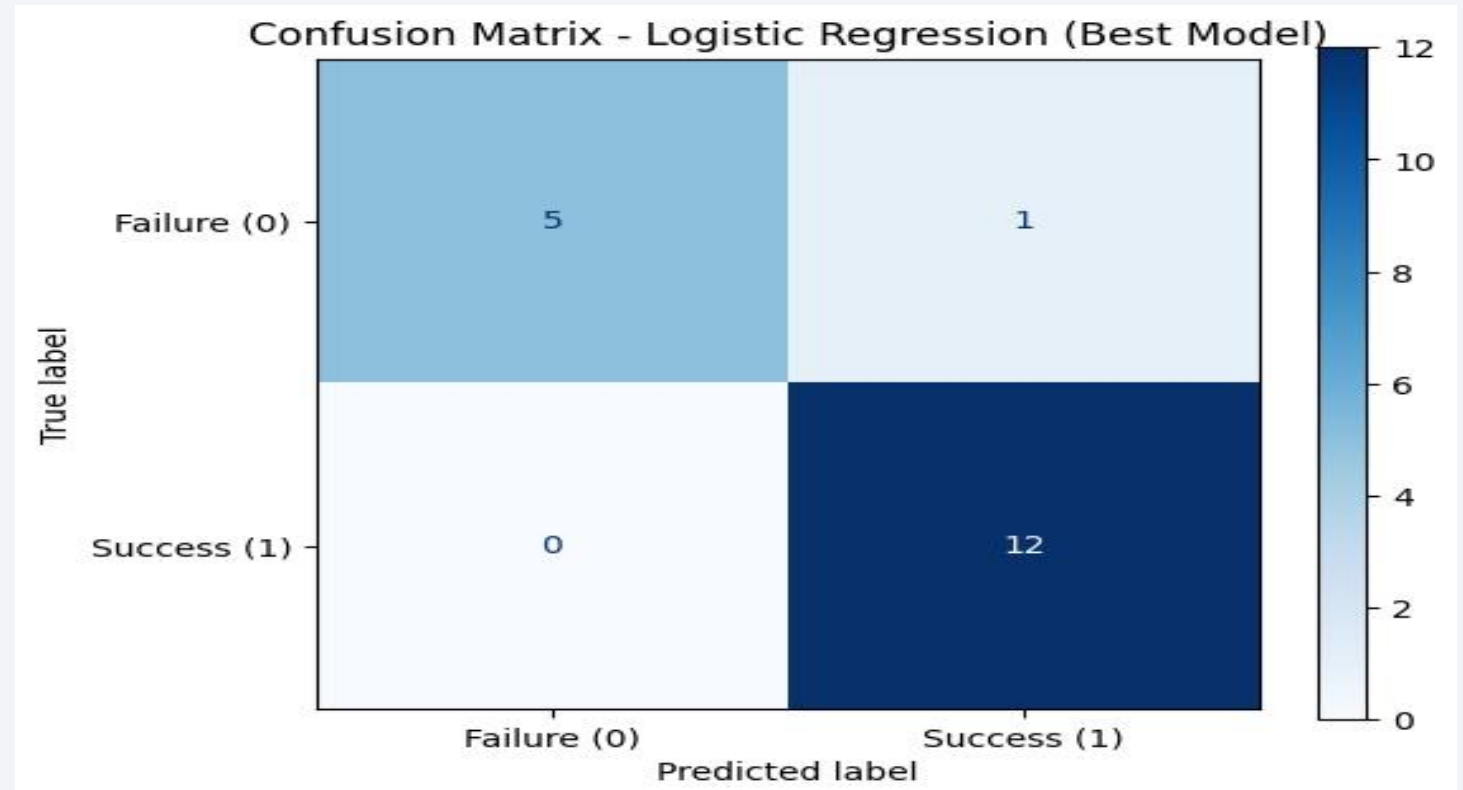
GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Confusion Matrix

Confusion Matrix - Logistic Regression (Best Model):

The confusion matrix shows the Logistic Regression model correctly predicted 5 failures and 12 successes, with only 1 false positive and zero false negatives. This gives the model 100% recall for successful landings and demonstrates excellent performance in identifying successful booster recoveries.



Conclusions

Point 1 - Data Collection & Processing Successfully Completed:

SpaceX launch data was successfully collected from two sources: the SpaceX API (107 records) and Wikipedia web scraping. Data wrangling transformed raw data into a clean dataset with a binary landing success label (Class: 1=success, 0=failure), achieving a 66.7% overall success rate across 90 Falcon 9 missions.

GitHub URL

<https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Conclusions

Point 2 - EDA Revealed Key Success Factors:

KSC LC-39A is the most successful launch site with 41.7% of all successful launches and a 76.9% success rate. The FT booster version achieved nearly 100% success, while GTO missions had the lowest success rate at 51.9%. The yearly trend shows dramatic improvement after 2015, with success rates reaching 100% in 2016, 2017, and 2019.

Conclusions

Point 3 - Geographic & Infrastructure Insights:

All four launch sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A in Florida; VAFB SLC-4E in California) are strategically located near coastlines (3-12 km) for safety and near highways/railways (0.4-1.1 km) for rocket component transport. Folium maps effectively visualized color-coded launch outcomes and proximity distances.

GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Conclusions

Point 4 - Machine Learning Successfully Predicts Landing Outcomes:

Point 4 - Machine Learning Successfully Predicts Landing Outcomes

Four classification models were trained and evaluated using GridSearchCV with 10-fold cross-validation. Logistic Regression, SVM, and KNN tied at 83.3% test accuracy. Logistic Regression was selected as the best model due to its simplicity and interpretability, achieving 100% recall in identifying successful landings with zero false negatives, proving that ML can effectively predict Falcon 9 first stage landing success.

GitHub URL

https://github.com/elias860/Applied-Data-Science-Capstone-Project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Appendix

Notebooks:

- SpaceX API Data Collection
- Wikipedia Web Scraping
- Data Wrangling & EDA
- EDA with SQL
- Machine Learning Prediction
- Folium Interactive Maps
- Dash Dashboard

Appendix

Key SQL Queries:

- SELECT DISTINCT launch sites
- SELECT * WHERE launch site LIKE 'CCA%' LIMIT 5
- SELECT SUM(PAYLOAD_MASS) WHERE Customer = 'NASA (CRS)'
- SELECT AVG(PAYLOAD_MASS) WHERE Booster_Version = 'F9 v1.1'
- SELECT MIN(Date) WHERE Landing_Outcome = 'Success (ground pad)'
- SELECT Booster_Version WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS BETWEEN 4000 AND 6000
- SELECT Mission_Outcome, COUNT(*) GROUP BY Mission_Outcome
- SELECT Booster_Version WHERE PAYLOAD_MASS = (SELECT MAX(...))
- SELECT month, Landing_Outcome, Booster_Version, Launch_Site WHERE year = 2015
- SELECT Landing_Outcome, COUNT(*) GROUP BY Landing_Outcome ORDER BY COUNT DESC

Appendix

Charts & Visualizations

- Launch site distribution bar chart
- Orbit type distribution bar chart
- Landing outcomes bar chart
- Success rate by orbit type bar chart
- Flight number vs launch site scatter plot
- Payload mass vs launch site scatter plot
- Flight number vs orbit type scatter plot
- Payload vs orbit type scatter plot
- Yearly success trend line chart

Appendix

Interactive Maps (Folium)

- Map 1: All launch sites location map (US view)
- Map 2: Color-coded success/failure markers map
- Map 3: Distance lines to coastline, highway, and railway

Key Python Libraries Used:

- requests, BeautifulSoup, pandas, numpy, matplotlib, seaborn, folium, dash, plotly, scikit-learn

Thank you!

